

Lecture Outline

- What is SDI?
- Sampling the Analogue Video Signal
- Basic block diagram of an SDI encoder
 - For SD and HD SDI, including:
 - Sampling, quantising, timing reference signals, ancillary data, serialising the data, wire encoding.
- Wire Encoding
- Luminance and Chrominance Sampling
 - 4:2:2, 4:1:1, 4:4:4, 4:2:0
- Levels
- Timing Reference Signals
- Eye display
- Ancillary Data
- Standards Overview
- Cables, Connectors and Distribution
- Introduction to Video Timing.

What is SDI?

Serial Digital Interface

The core idea is to create a colour television signal that is also **backward compatible** with older monochrome (black & white) televisions. This is achieved by separating the picture information into two main components: **Luminance** and **Chrominance**.

Broadcaster

Viewer

$$Y' = 0.21R' + 0.72G' + 0.07B'$$

HD video, to 2 decimal places

$$Y' = 0.3R' + 0.59G' + 0.11B'$$

SD video, to 2 decimal places

Y' signal alone can be displayed on a monochrome TV to produce a black-and-white image.

Colour Camera
(analogue colour television signals)

Colour Splitter

Gamma Corrector

Y

Y' - Luminance

C - Chrominance
(Colour Difference Signals)

Colour Coding

- PAL
- NTSC
- SECAM

Monochrome TV
(Black & White)



PAL /
NTSC /
SECAM
Decoder

Colour TV
Display

Colour TV

To represent the colour information efficiently, "colour difference" signals are created by subtracting the luminance (Y') from the red (R') and blue (B') signals

Inside a PAL Coder, R'-Y' and B'-Y' are rescaled and called u and v.

$$u = 0.493(B' - Y')$$

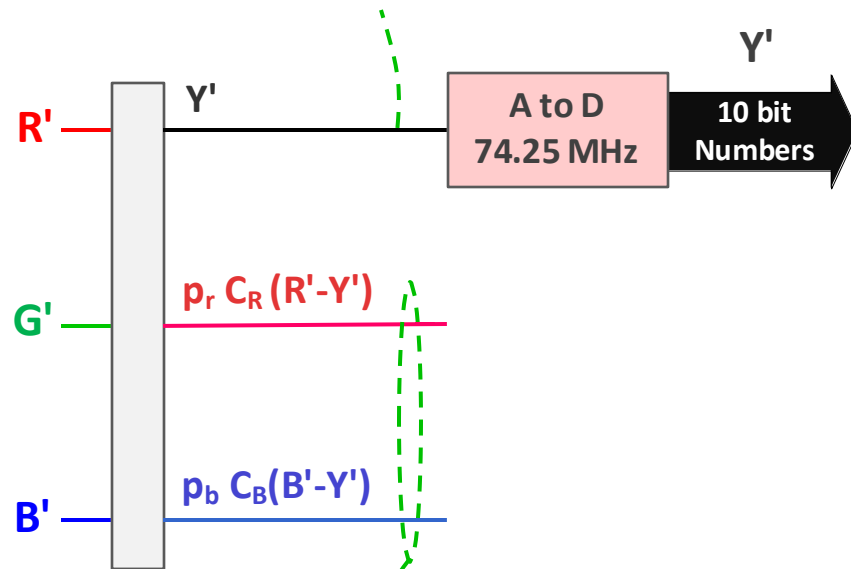
$$v = 0.877(R' - Y')$$

Camera captures the scene, and internally, the light is split into its primary colour components: Red (R), Green (G), and Blue (B)



Y – Luma

The black and white (monochrome) picture



C – Chrominance

The colouredness of the picture

p_r and p_b (aka C_R and C_B) are scaled version of, $R'-Y'$ and $B'-Y'$.

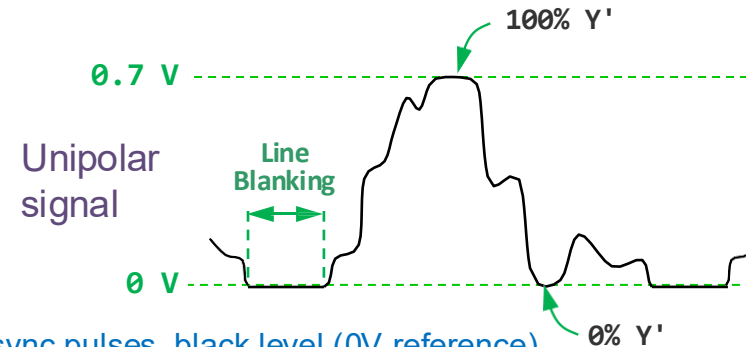
$$C_R = (B' - Y') / 1.8666$$

$$C_B = (R' - Y') / 1.5748$$

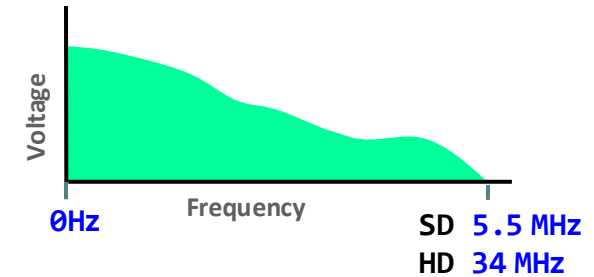


The Y' signal looks something like this on an oscilloscope or waveform monitor:

Reference levels like 0% (Black Level - the reference for darkest parts of the picture) and 100% Y' (White Level - the reference for the brightest parts)



...and something like this on a spectrum analyzer:



sync pulses, black level (0V reference)

- **Core of digitization** is converting the continuous analogue signals (like Y') into a sequence of discrete digital numbers. This involves two main steps:

Sampling:

- The rule defined by **Harry Nyquist** says that we must sample **at least twice** the **highest frequency** in the signal. So for **HD** that is at least **68MHz**.
- The standard (ITU-R BT.709) states that we sample at **74.25MHz** (74.25 million times/s).

Quantization:

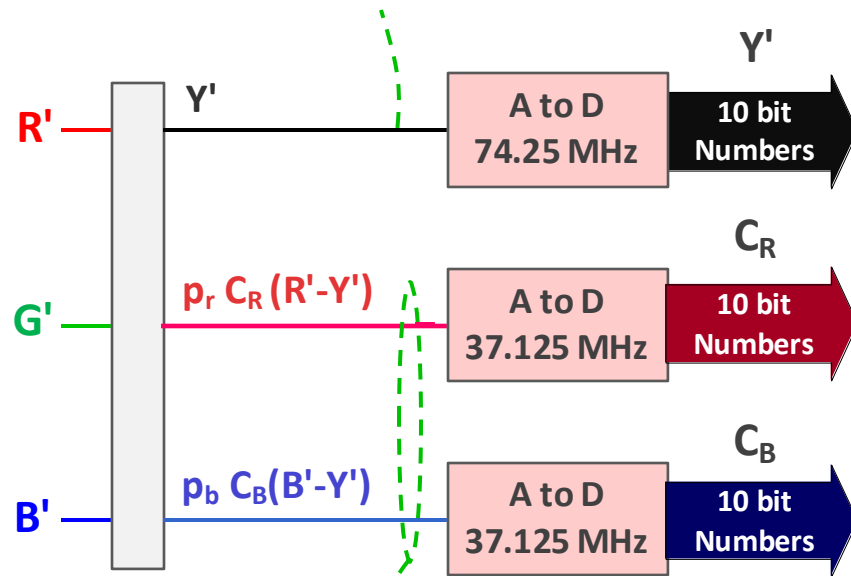
- For each sample taken, the ADC assigns a discrete numerical value corresponding to the measured voltage.
- Each **sample** is stored/transmitted as a **10 bit** binary number. This means each sample is represented by a 10-bit binary number, allowing for $2^{10}=1024$ distinct levels to represent the signal's amplitude range (from black to white for Y').

Separation of the video signal into Luma (Y') and Chrominance (CB, CR) and their digitization.



Y – Luma

The black and white (monochrome) picture



C – Chrominance

The colouredness of the picture

p_r and p_b (aka C_R and C_B) are scaled version of, $R' - Y'$ and $B' - Y'$.

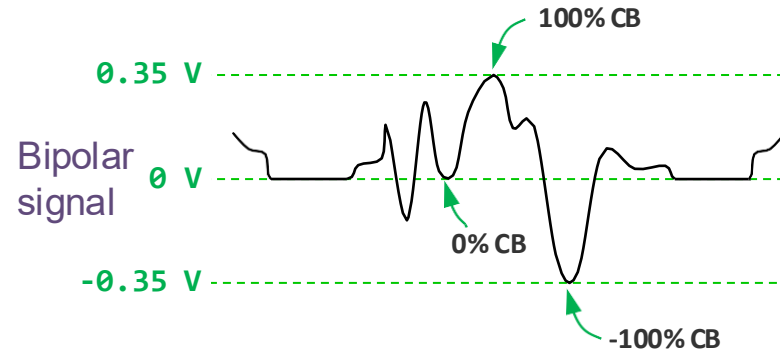
$$C_R = (B' - Y') / 1.8666$$

$$C_B = (R' - Y') / 1.5748$$

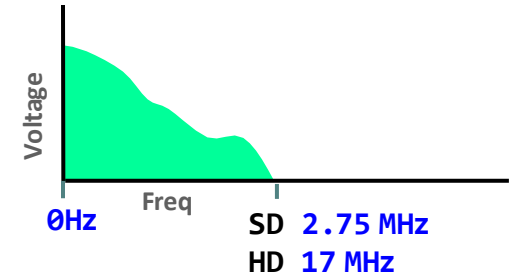


The CR or CB signals look something like this on an oscilloscope or waveform monitor: representing the colour information relative to luma

For **CB**, the positive peak (e.g., +100% CB) corresponds to the most saturated **blue** colour, while the negative peak (e.g., -100% CB) corresponds to the most saturated **yellow**.



...and something like this on a spectrum analyzer:



For **CR**, the positive peak (+100% CR) corresponds to the most saturated **red**, and the negative peak (-100% CR) corresponds to the most saturated **cyan**.

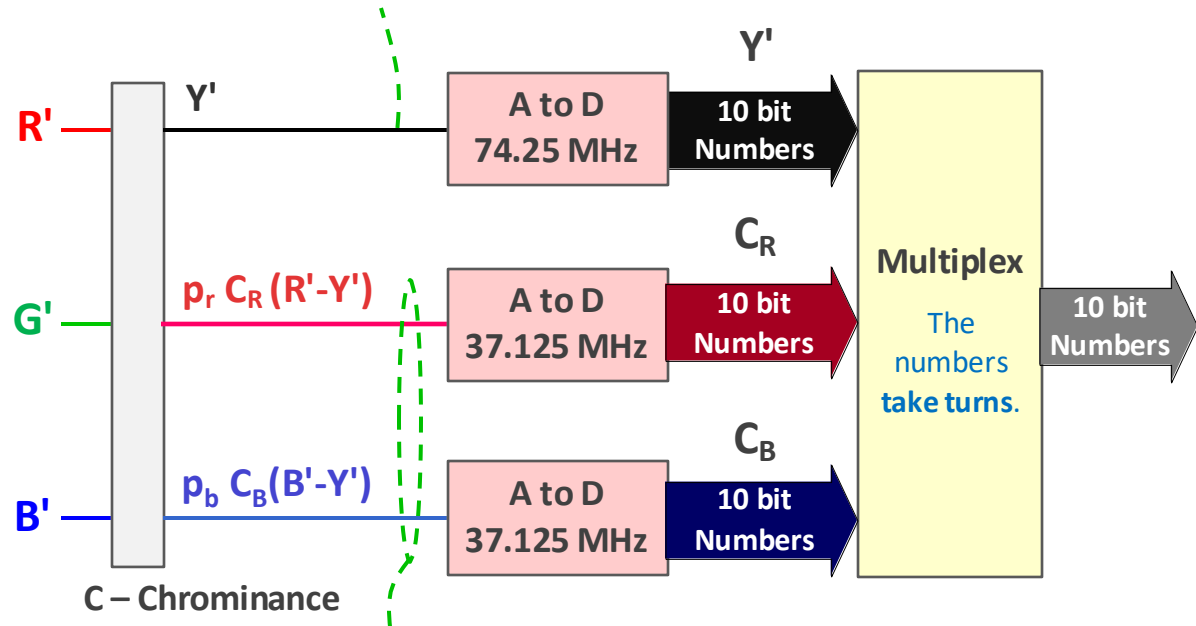
0% CB (or 0% CR): When the CB or CR signal is at this 0% level, it means there is **no colour difference** from the luma (Y') component at that point in the image.

- The rule defined by **Harry Nyquist** says that we must sample **at least twice** the **highest frequency** in the signal. So for **HD** that is at least **34Mhz**.
- The standard (**ITU-R BT.709**) states that we sample at **37.125Mhz**.
- Each **sample** is stored/transmitted as a **10 bit** binary number.

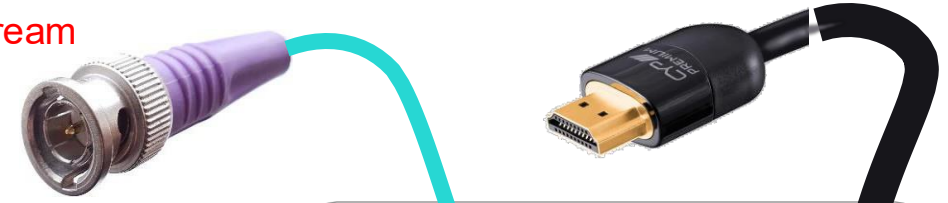


Y – Luma

The black and white (monochrome) picture



From the multiplexed 10-bit digital stream (output), the video data can be sent through various digital interfaces or processed further:



SDI (Serial Digital Interface)

HDMI (High-Definition Multimedia Interface)

Serialise (bits take turns) as voltages on a wire (or wires)

Packetize and put onto an IP Network

SMPTE 2022-6 (inc. audio + data)
SMPTE 2110-20 (video only)

compress
JPEG-XS etc.

Packetize and put onto an IP Network

SMPTE 2110-22

comp
MPEG

Packetize to MPEG Transport Stream

SDI Voltages & Data Rate

ASI
(Asynchronous Serial Interface)

IP Packets

SMPTE2022-2/3/4
(TSolP, MPEGoIP)

save as data in a file

MOV

MXF

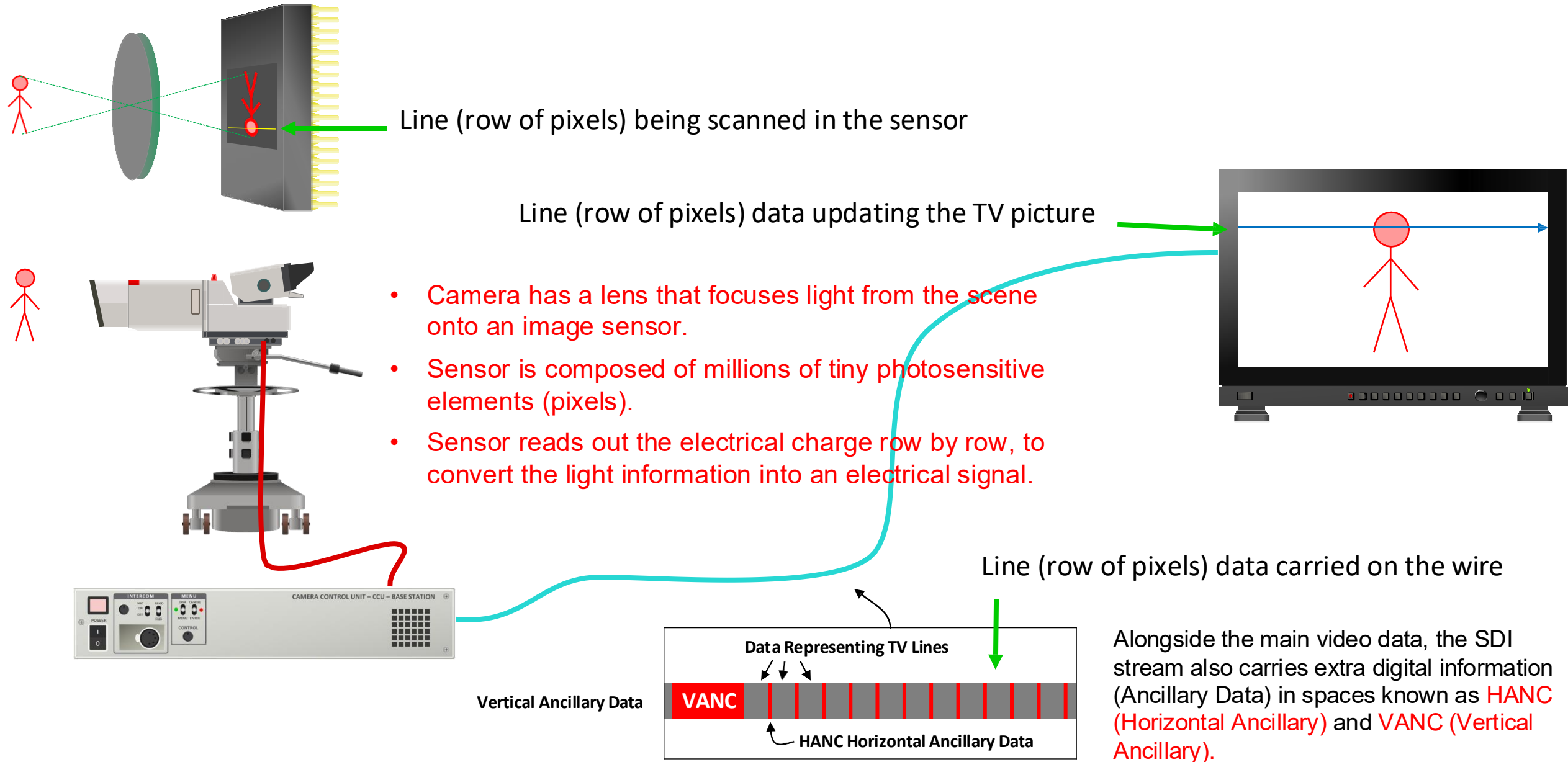
MP4 etc...

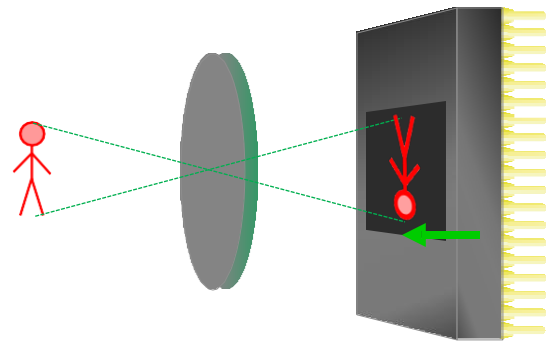
compress
MPEG etc.

save as data in a file



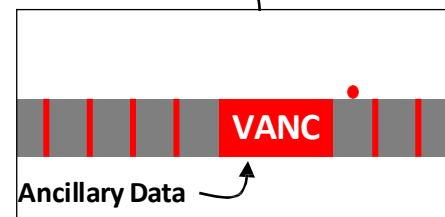
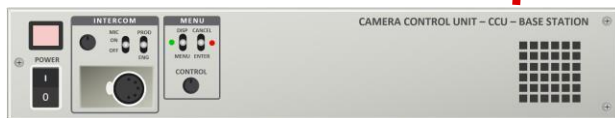
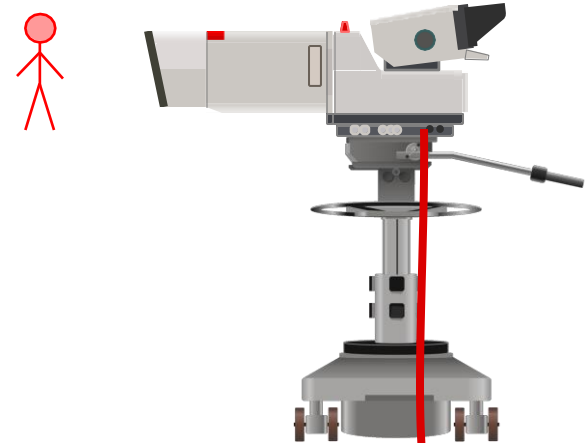
- Separate streams of Y, Cb, and Cr 10-bit numbers need to be combined into a single serial data stream for easier transmission.
- Multiplexing is the process where the numbers from each component "take turns" being sent in a defined order.
- For example, samples might be interleaved in a sequence like Y, Cb, Y, Cr, Y, Cb, Y, Cr, and so on (this is characteristic of 4:2:2 chroma subsampling).
- This creates a single stream of 10-bit numbers representing the complete digital video signal.





Within the SDI digital stream, in the vertical blanking interval (the time between video frames), there is dedicated space for carrying **Vertical Ancillary Data**.

VANC space is used to embed various types of non-picture information, such as audio, timecode, closed captions, and metadata, alongside the video signal itself.



SD – 270 Mbits / sec

- SMPTE ST 259M
- ITU-R BT.601
- ITU-R BT.656

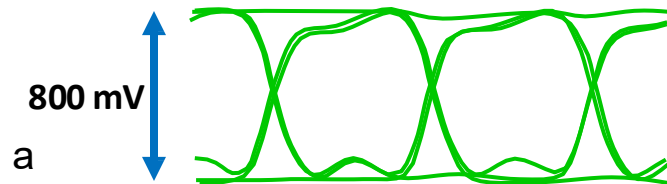
HD – 1.485 Gbits / sec

- SMPTE ST 292M
- ITU-R BT.709
- ITU-R BT.1120

UHD – 12 Gbits / sec

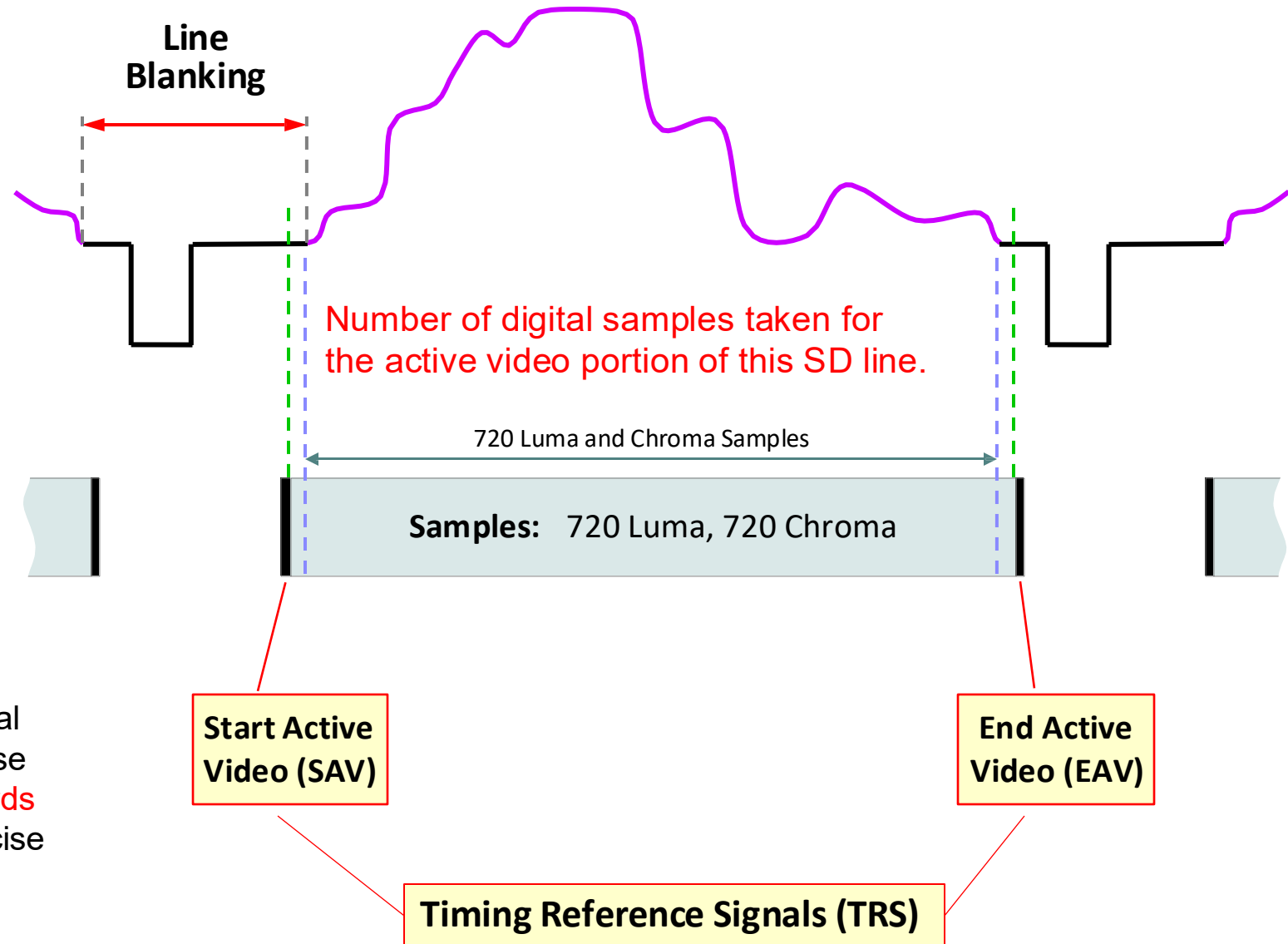
- SMPTE ST 2082
- ITU-R BT.2020
- ITU-R ITU-R BT.2077

- **ITU-R BT.601** is a standard for digitizing analog interlaced SD video signals, defining parameters like sampling structure (e.g., 4:2:2) and levels.
- **ITU-R BT.656** is a standard for the parallel interface carrying digital component video signals
- **ITU-R BT.709** is the fundamental standard for High-Definition Television, defining aspects like image resolution (e.g., 1920x1080, 1280x720), frame rates, colour space, and the relationship between R'G'B' and Y'CbCr.
- **ITU-R BT.1120** specifies the digital interfaces for HDTV studio applications.
- **ITU-R BT.2020** is a standard defining parameters for UHD TV, including resolution (3840x2160, 7680x4320), frame rates, and a much wider colour space than Rec. 709.
- **ITU-R BT.2077** specifies the digital interfaces for UHD TV studio applications.

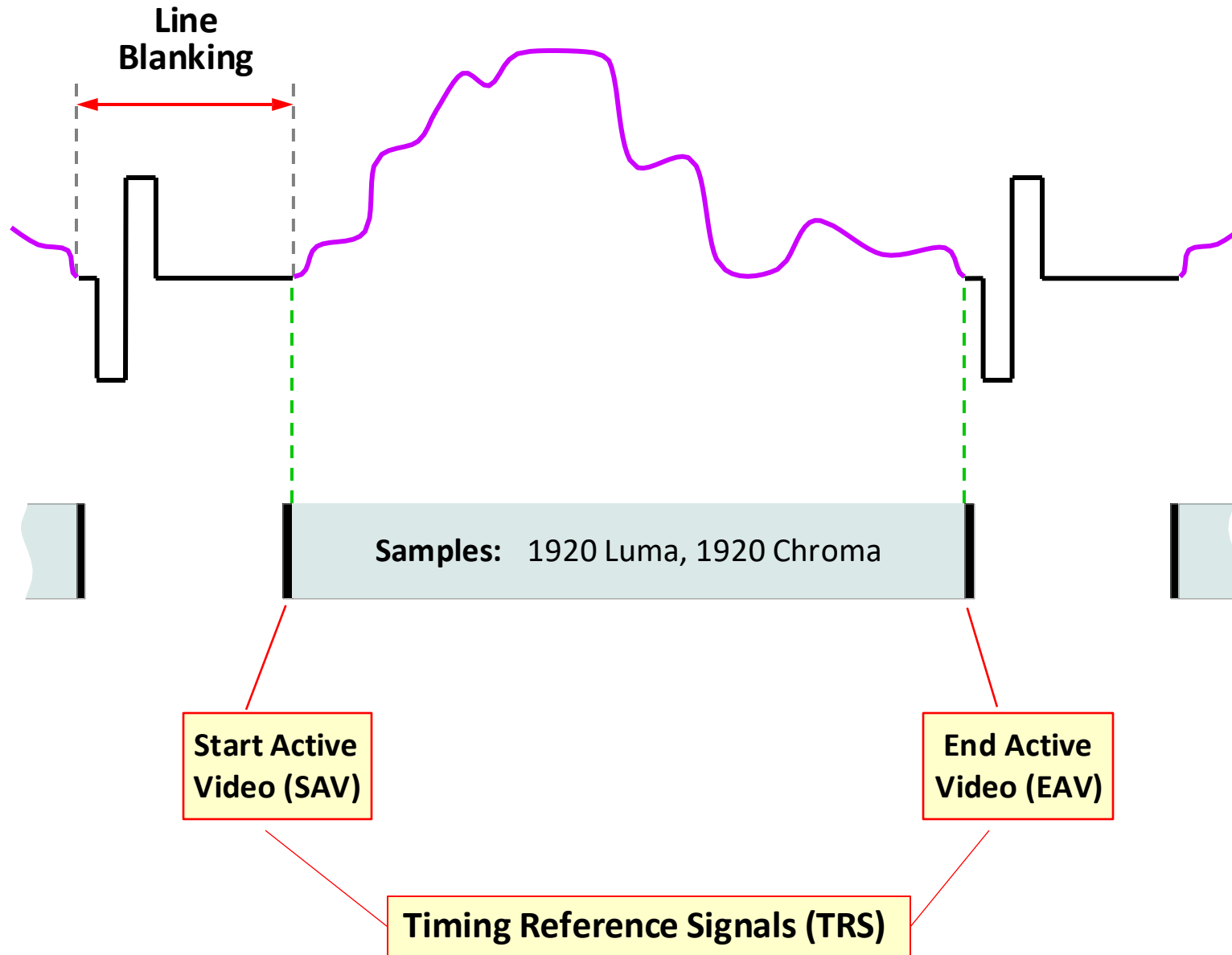


SDI – Serial Digital Interface

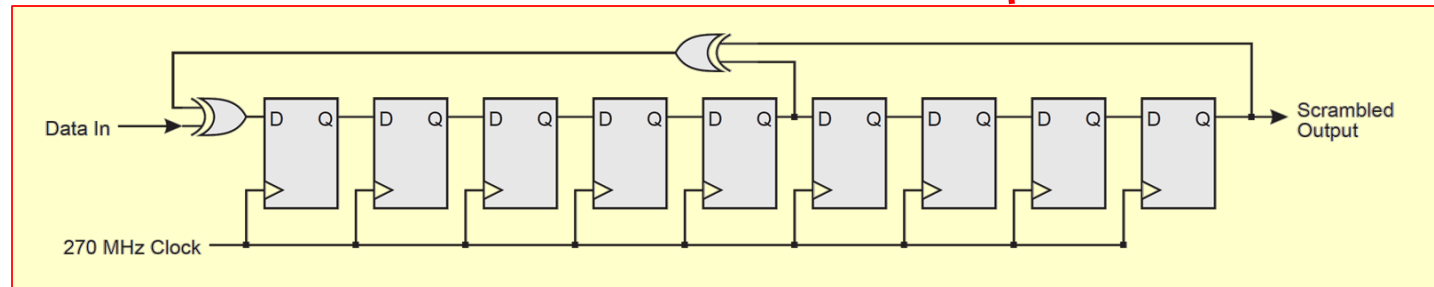
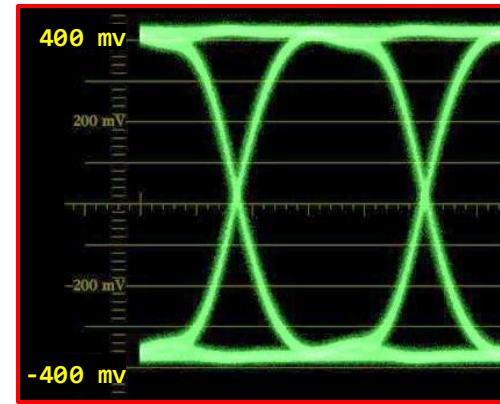
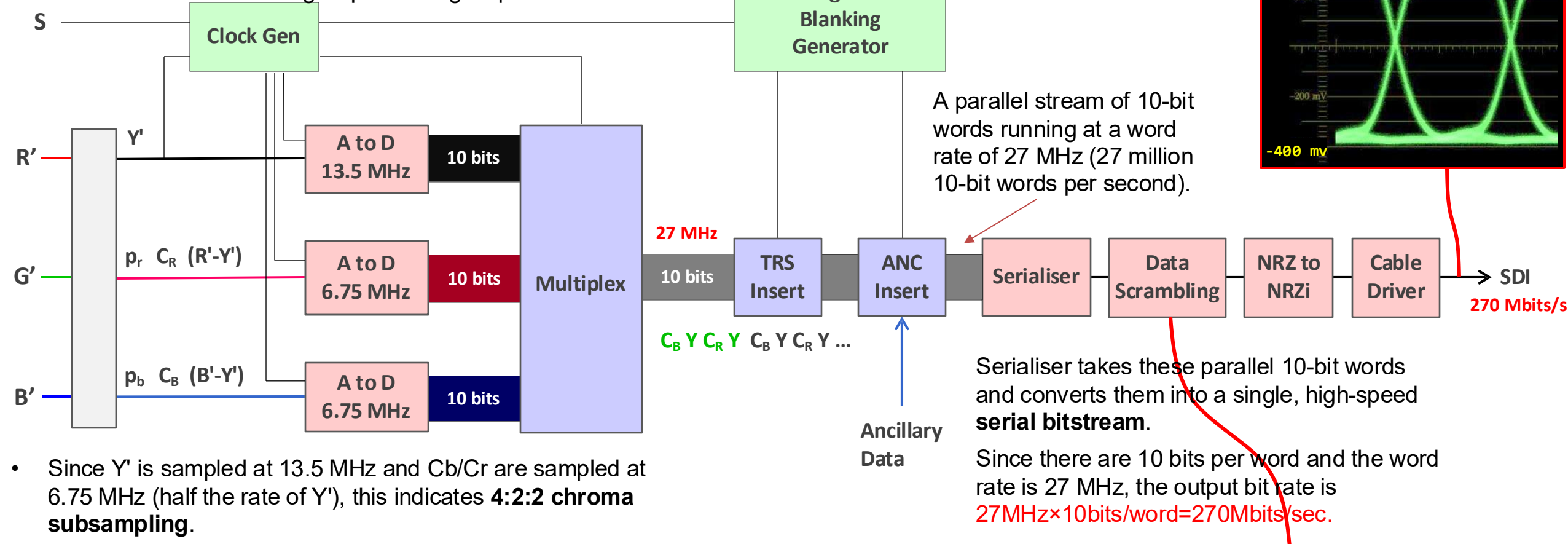
Line Blanking: This is the interval of time between the **end of one active video line** and the **start of the next active video line** on the same frame or field. During this period, no picture information is transmitted.



Timing Reference Signals (TRS): Embedded within the Line Blanking interval are critical digital signals called TRS. These are specific, **recognizable digital code words** that the receiving equipment uses for precise **synchronization**.



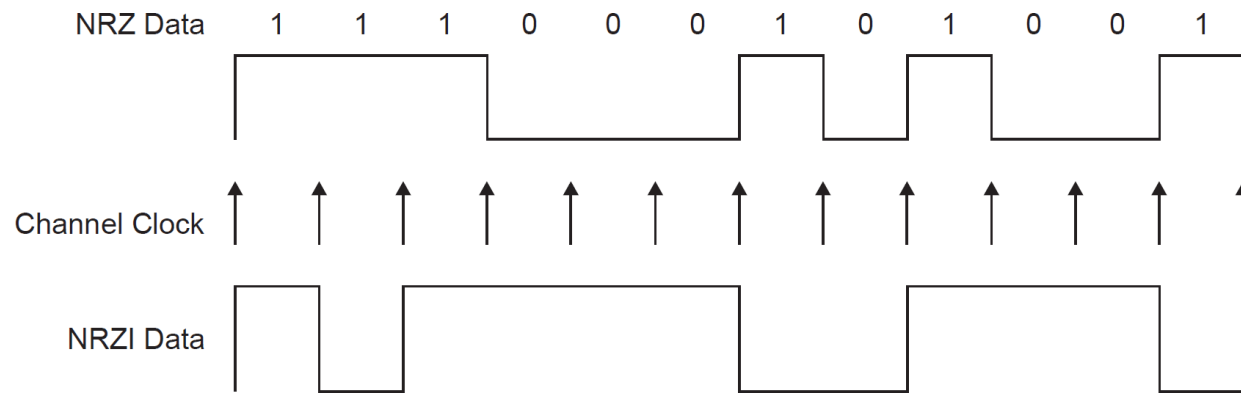
Clock Generator: It takes the sync information from the input (S) and generates all the precise clock frequencies needed for the various digital processing steps.



- Since Y' is sampled at 13.5 MHz and Cb/Cr are sampled at 6.75 MHz (half the rate of Y'), this indicates **4:2:2 chroma subsampling**.
- **Multiplexer** uses the 27 MHz clock to output of these interleaved 10-bit words.
- **TRS Insert** takes the multiplexed Y'CbCr data and inserts the **Timing Reference Signals (TRS)** generated by the Digital Blanking Generator.

- SDI uses **NRZi** encoding.

- Data = **1** **change** at clock edge.
- Data = **0** **do not change** at clock edge.

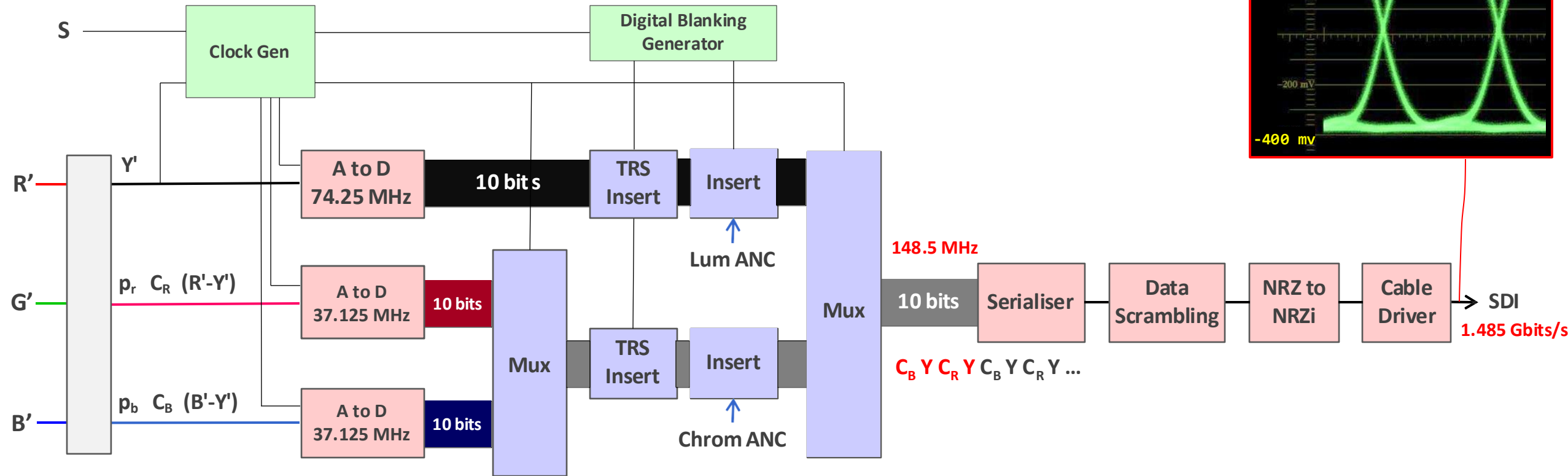


NRZ = Non-Return-to-Zero
0 and **1** represented by **two different voltages** (one of which could be zero volts).
 As opposed to **Return-to-Zero** where the voltage always goes back to **0V** for a while between bits.
Return to Zero is self-clocking. **NRZ** is not.
NRZi (i for inverted) is where **0** is constant and **1** is toggle (or the other way round). *The name does not really make much sense...*

RZ guarantees a transition within each bit period, which is useful for clock recovery, but it requires more bandwidth than NRZ/NRZI as there are more transitions per bit.

- A long run of **0s** means **no change of voltage** – this is not good because there is too much low frequency energy.
 - If the signal stays at a constant voltage for a long time (due to many consecutive 0s), the receiver's clock recovery circuit loses its reference and can drift out of sync, leading to errors in decoding the data.

So we multiply (XOR) the data by **self-clocking pseudo-random sequence**. This significantly reduces the probability of having very long runs of either 0s or 1s, even if the original data had them.



- Unlike the single multiplexer in the SD diagram, the HD process often involves a first multiplexing stage specifically for the chrominance components.
- The 10-bit C_b and C_r streams, both at a 37.125 MHz sample rate, are multiplexed together. This creates a single stream of 10-bit words at a rate of $37.125 \text{ MHz} \times 2 = 74.25 \text{ MHz}$.
- The diagram shows the multiplexer operating with a 148.5 MHz clock. This clock rate corresponds to the word rate of the final 10-bit stream ($1.485 \text{ Gbits/sec} / 10 \text{ bits/word} = 148.5 \text{ Mwords/sec}$).
- The output bit rate is $148.5 \text{ Mwords/sec} \times 10 \text{ bits/word} = 1485 \text{ Mbits/sec} = 1.485 \text{ Gbits/sec}$.

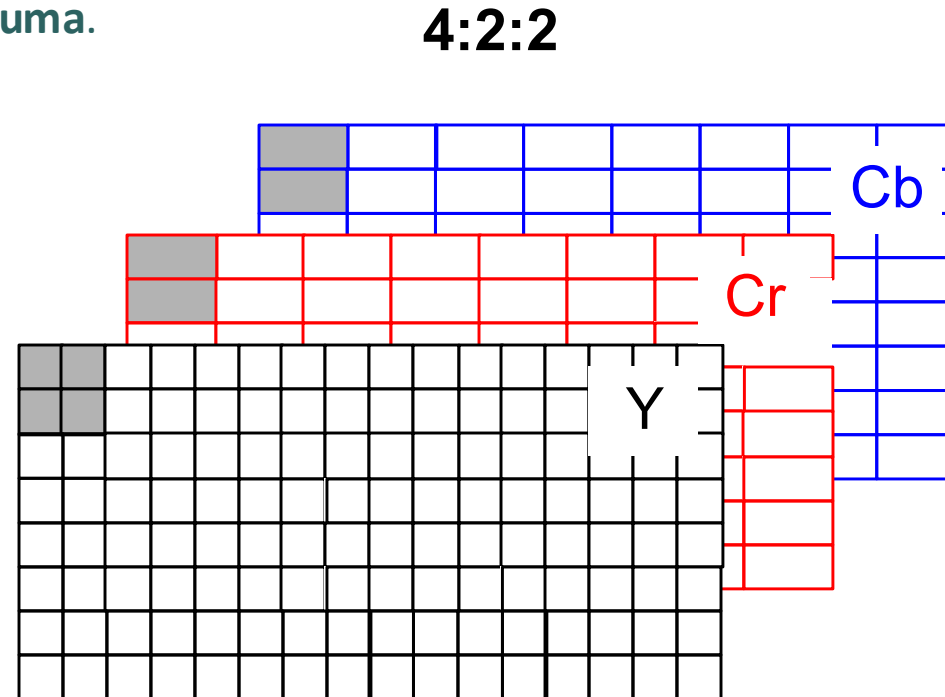
- The **sample rate** is a **multiple of line frequency**.
- **Human vision is significantly more sensitive to detail in brightness (luma) than in colour (chroma)**. Therefore, to reduce the amount of data required to represent the video, the chrominance components (Cb and Cr) are typically sampled at a lower spatial resolution (meaning fewer samples are taken) than the luma component. – This is called **chroma subsampling**.
- Common chroma subsampling scheme are:
 - **4:2:2**
 - **4:4:4**
 - **4:2:0**
 - **4:1:1**
- Originally, each number represented the **ratios of sample rates**.
 - **4:2:2** meaning that the luminance (**4**) was sampled twice of often as Cb (**2**) and Cr (**2**).
- The alternative is called **J:a:b** notation:
 - **J**: The horizontal sampling reference (the width in **Y** pixels of a conceptual region). Usually, **4**.
 - **a**: number of chrominance samples (**Cr**, **Cb**) in the first row of J pixels.
 - **b**: number of changes of chrominance samples (Cr, Cb) between first and second row of J pixels.

- Horizontal chroma resolution is half that of luma.

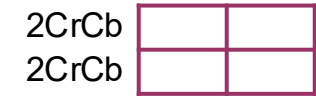
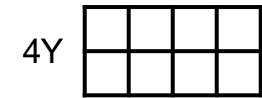
Each chroma sample horizontally spans the width of two luma samples. This illustrates the reduced horizontal resolution for chroma.

- Vertical resolutions are the same.

Cb and Cr samples align vertically, and there are chroma samples on every horizontal line shown, demonstrating that there is no vertical subsampling.



J:a:b notation

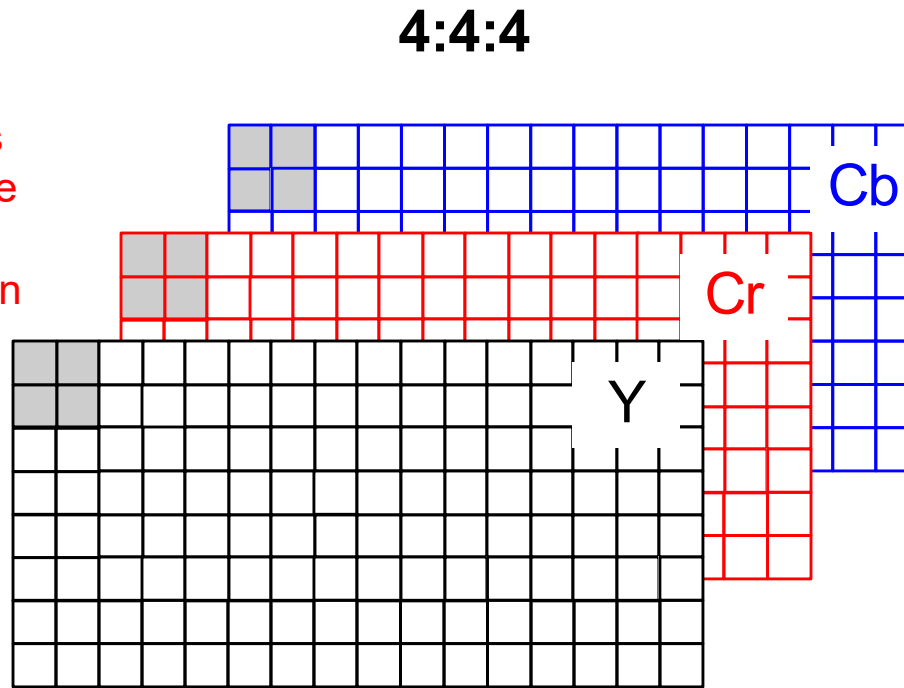


For 4 Luma samples horizontally, there are typically 2 Chrominance samples (meaning 2 Cb samples and 2 Cr samples) in the first row, and 2 Chrominance samples in the second row

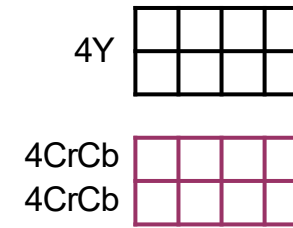
- Used in standard SDI (SD and HD). Most broadcast centre signals use it.
- Video tape formats:** AVC-Intra 100, Digital Betacam, Betacam SX
- Codecs:** ProRes (HQ, 422, LT, and Proxy), XDCAM HD422

- **Luma** and **chroma** have **equal resolutions** horizontally and vertically.

For 4 Luma samples horizontally, there are 4 Chrominance samples (meaning 4 Cb samples and 4 Cr samples) in the first row, and 4 Chrominance samples in the second row.



J:a:b notation



Chrominance is sampled at a 1:1 ratio with luma, both horizontally across each line and vertically down the image across multiple lines.

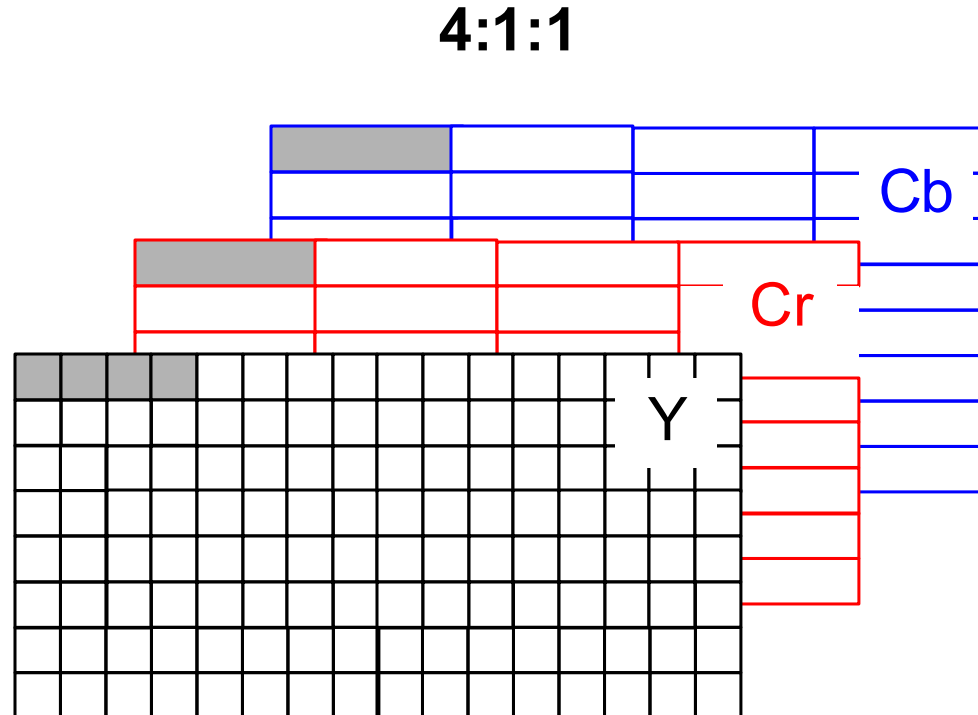
- 4:4:4 is used for high end graphics production, high end TV production and digital cinema. Because 4:4:4 retains the maximum amount of colour information, it is essential for applications where colour fidelity is paramount.
- HD 4:4:4 can be carried by dual link HD-SDI or 3G-SDI. To transmit 4:4:4 HD at the original 1.485 Gbit/s rate, two standard 1.485 Gbit/s HD-SDI cables are used simultaneously. One link typically carries the Y component, while the second link carries the Cb and Cr components.

- **Horizontal chroma resolution is a quarter** that of luma.

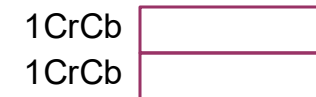
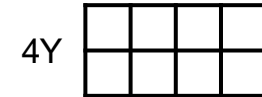
For 4 Luma samples horizontally, there is typically **1** Chrominance sample (meaning 1 Cb sample and 1 Cr sample) in the first row, and **1** Chrominance sample in the second row.

- **Vertical resolutions are the same.**

Because you only take 1 chroma sample for every 4 luma samples horizontally, the level of detail captured in the colour information across the width of the picture is reduced to one-fourth compared to the brightness information.



J:a:b notation



- **Very bad** for graphics involving a chroma key.

➤ Although it provides good data reduction and was used in various tape formats, its low horizontal chroma detail makes it less suitable for tasks requiring precise colour information, such as high-quality chroma keying.

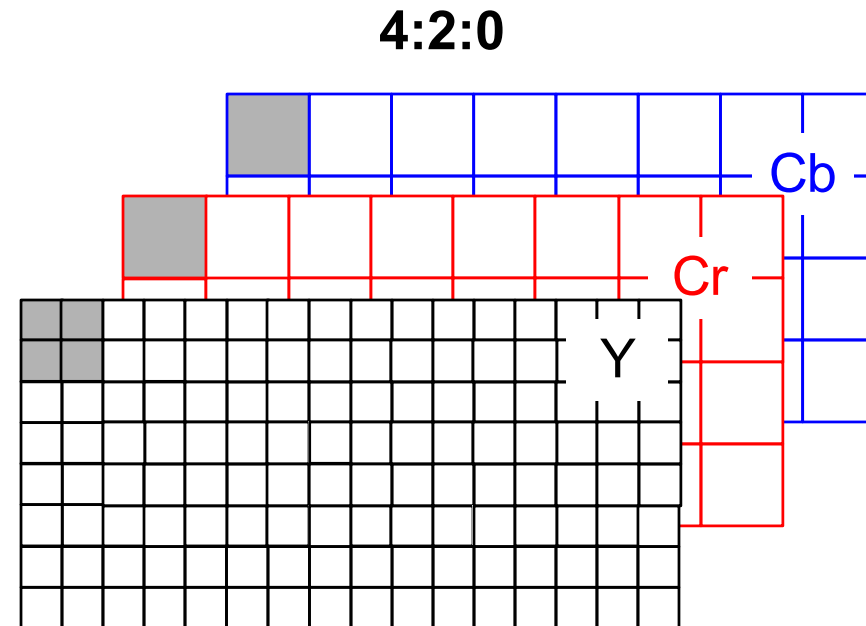
- **Video tape formats:** DVCPRO, 480i ("NTSC") DV and DVCAM

- **Horizontal and vertical chroma resolution is half that of luma.**

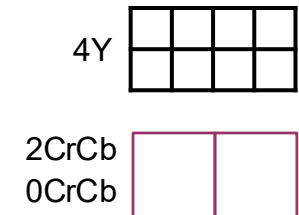
- **For 4 Luma samples horizontally:**

- There are 2 Chrominance samples (meaning 2 Cb samples and 2 Cr samples) in the first row. This indicates horizontal subsampling by half.
- There are 0 additional Chrominance samples in the second row relative to the first row's pattern. This indicates vertical subsampling, where chrominance information is not sampled on every line. The chroma information for a 2x2 block of Luma pixels is represented by a single chroma sample in this scheme.

A single chroma sample in 4:2:0 is considered to represent the colour information for a 2x2 block of luma pixels.

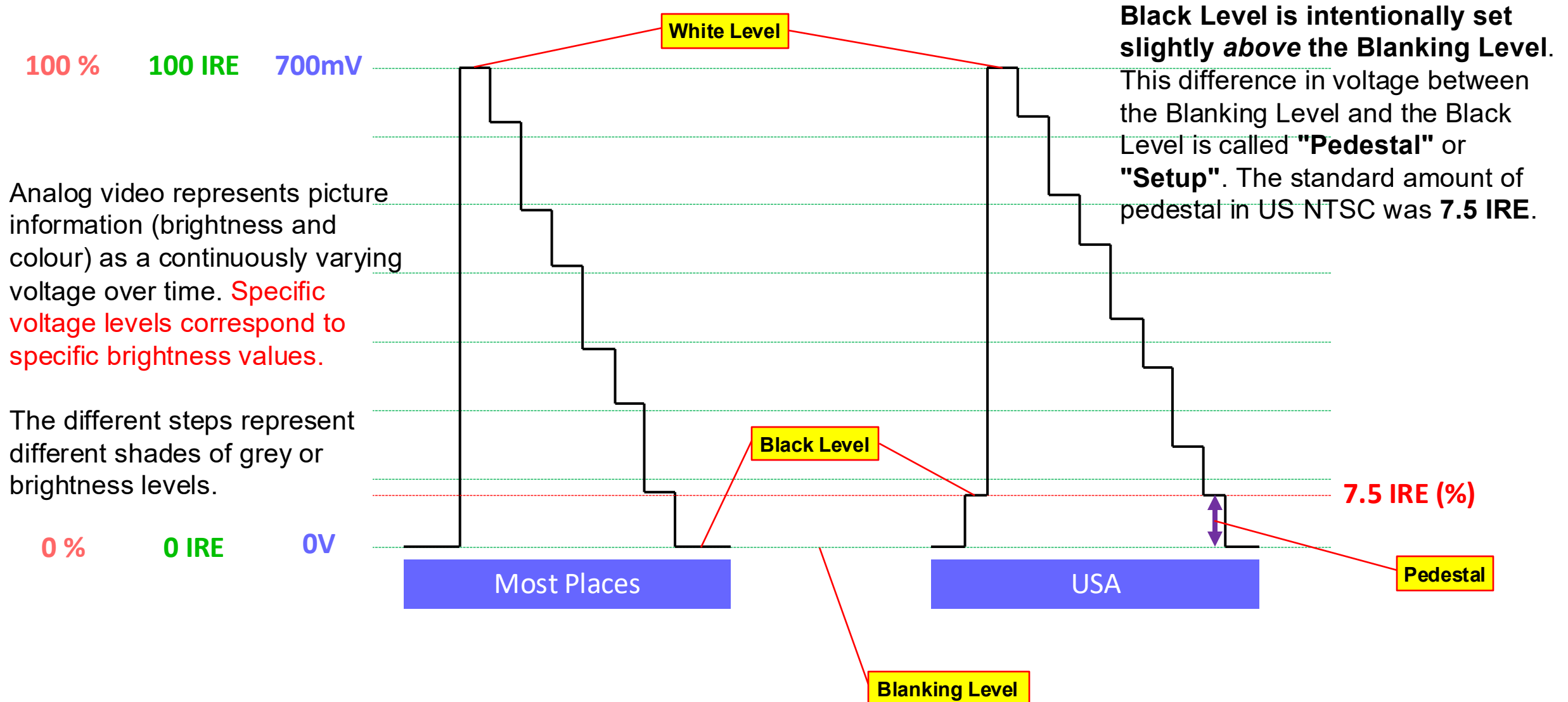


J:a:b notation



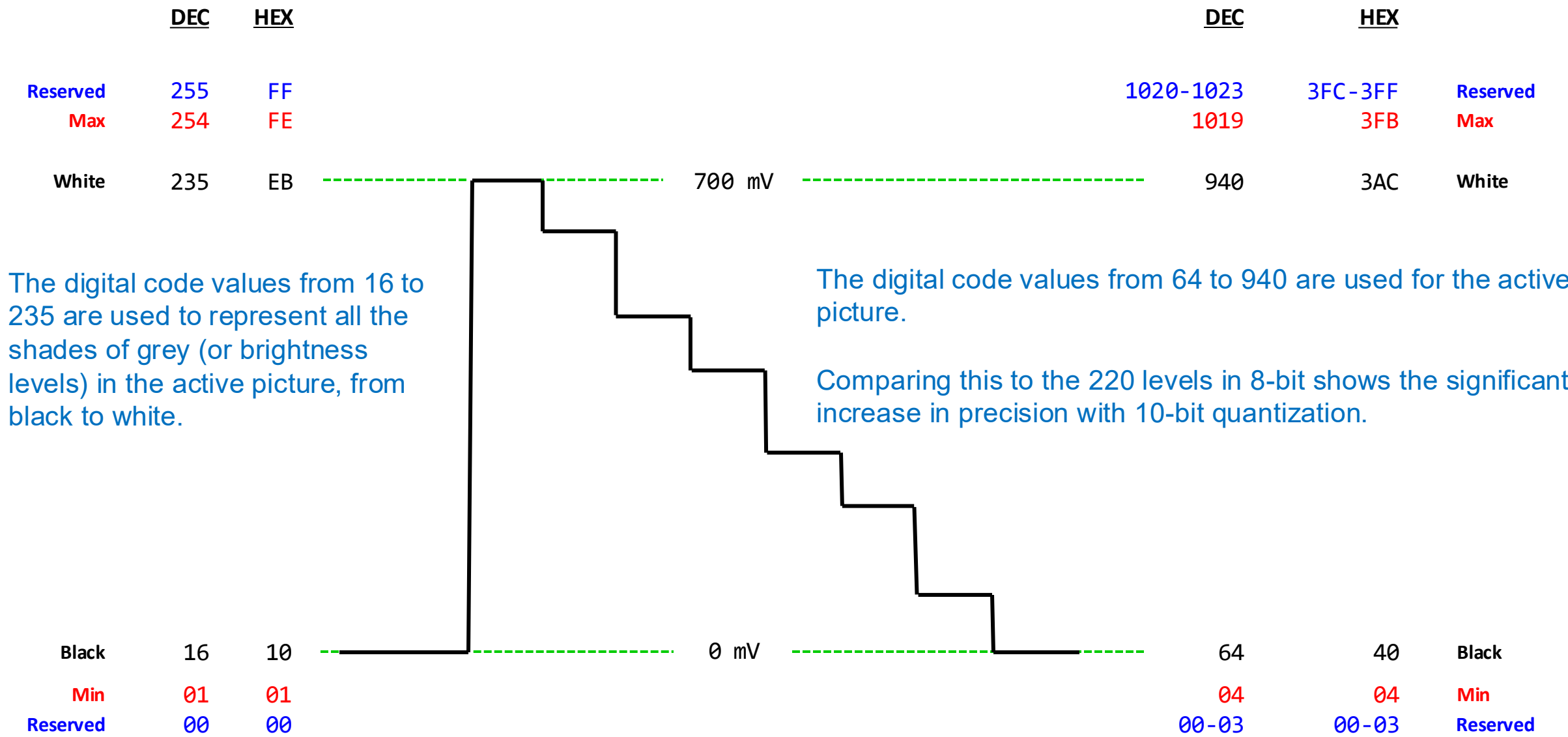
- **Video tape formats:** 576i ("PAL") DV and DVCAM

- **Codecs:** MPEG-2 part 2, MPEG-4 part 10/AVC, DVD-Video and Blu-ray, AVCHD, most JPEG

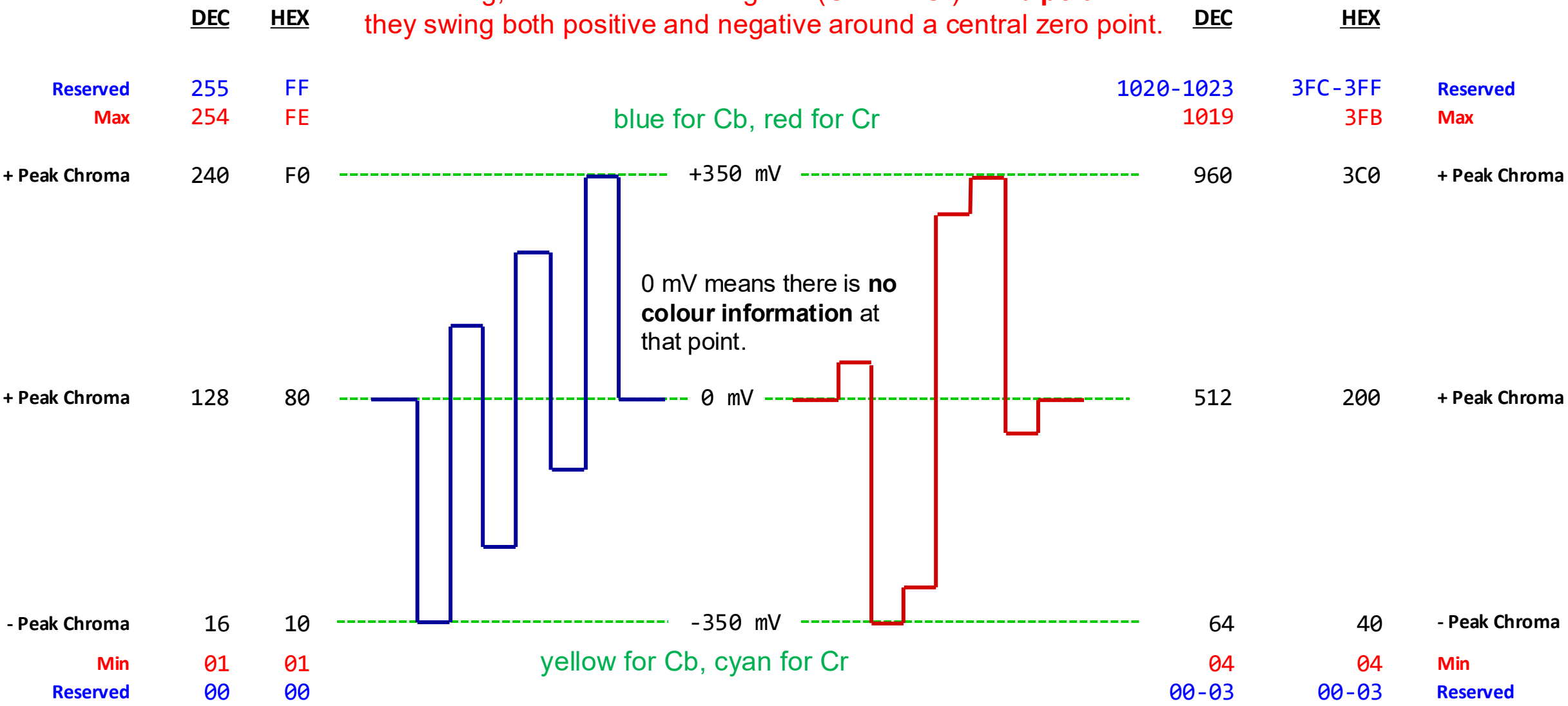


With 8-bit quantization, there are $2^8=256$ possible digital code values, ranging from 0 to 255.

With 10-bit quantization, there are $2^{10}=1024$ possible digital code values, ranging from 0 to 1023.



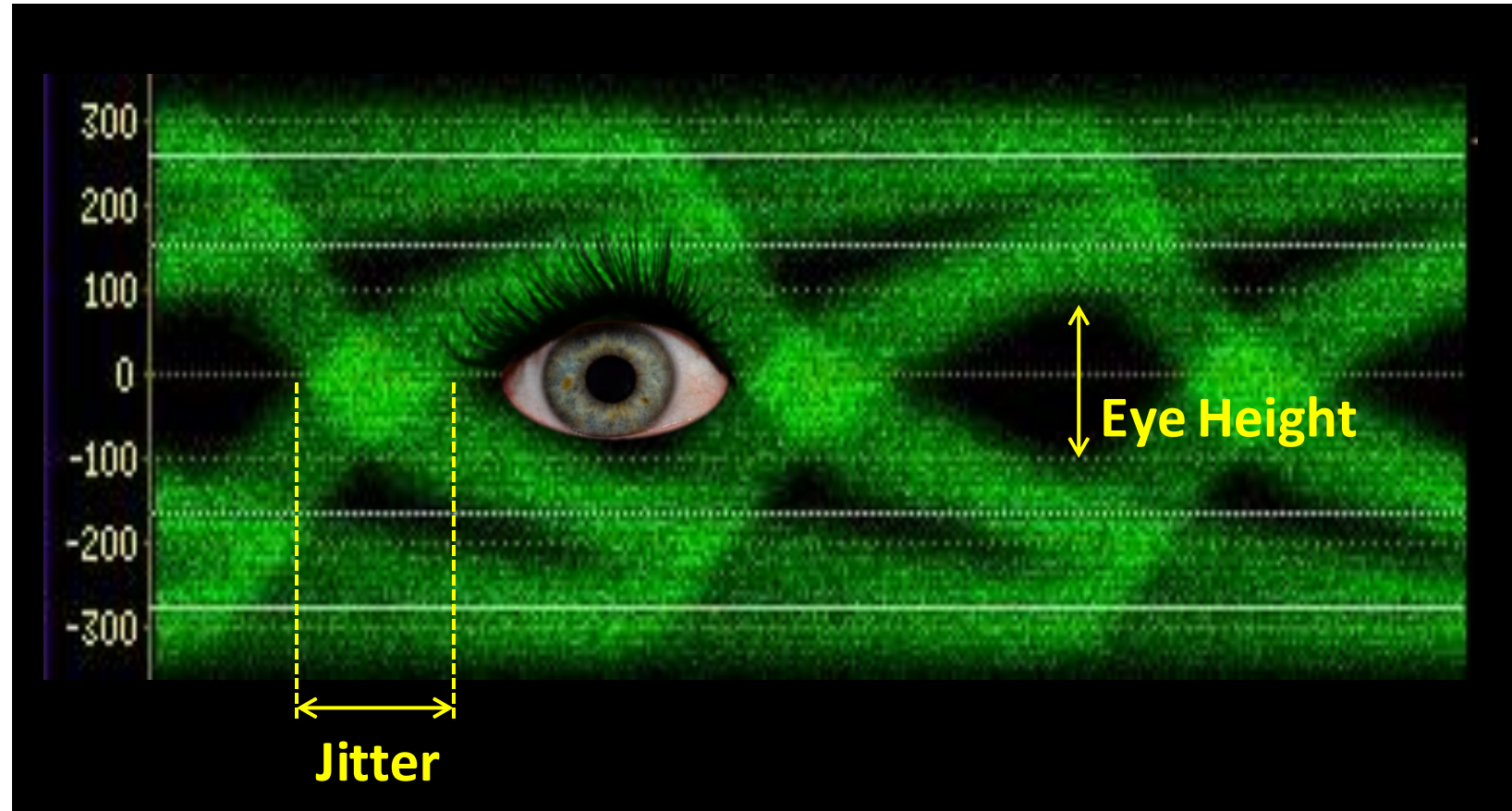
Unlike Luma, which is primarily a positive voltage signal relative to blanking, the chrominance signals (Cb and Cr) are **bipolar** – they swing both positive and negative around a central zero point.



- The **Timing Reference Signal** (TRS) provides synchronising information.
- It does a **similar job** as an **analogue sync pulse**.
- The **TRS** consist of **four 10-bit words**:
 - The **first 3 words** are always the same - In hex: 3FF,0,0 which in decimal is: 1023,0,0
 - The first three words of every TRS sequence have a fixed, constant value. These are often referred to as the TRS preamble or alignment sequence.
 - These levels **do not occur during active video** – so the receiver can **recognise** the TRS.
 - The **4th word** (**XYZ word**) consists of **3 flag bits**, along with an error correcting code.
 - See the next slide for details of this...
- There is a **TRS** at **both ends** of the **active line**.
 - **SAV** – Start Active Video (This is the signal that tells the receiver, "The picture data is about to begin.")
 - **EAV** – End Active Video (This tells the receiver, "The picture data for this line has ended.")

- The **flags bits** in the **4th word** are known as **H**, **F**, and **V**, contains crucial flag bits that provide essential synchronization and format information.
- The TRS at the **end** of the **active line** (therefore at the **beginning** of **horizontal blanking**) has the **H bit** set to **1**.
 - This is **End of Active Video**, or **EAV**.
- The TRS at the **start** of the **active line** (therefore at the **end** of **horizontal blanking**) has the **H bit** set to **0**;
 - This is the or **SAV**.
- The **V bit** is used to indicate the **start** of the **vertical blanking region**.
 - An **EAV** packet with **V bit** set to **1** indicates the **following line** is part of the **vertical interval**.
 - An **EAV** packet with **V bit** set to **0** indicates the **following line** is part of the **active picture**.
- The **F bit** is used in **interlaced** and **segmented-frame** formats.
 - **F bit** set to **0** means **field 1** or **segment 1**.
 - **F bit** set to **1** means **field 2** or **segment 2**.
 - In **progressive** scan formats, where frames are not divided into fields, the **F bit** is always set to **0**.

- Eye display shows the **voltage on the wire** – triggered on the clock edges.
 - **Eye height** represents the signal's amplitude margin.
 - The display also gives an indication of Jitter (delay variation).
-
- Jitter can be caused by various factors in the transmission path, such as reflections, interference, or imperfections in the clock generation circuits.



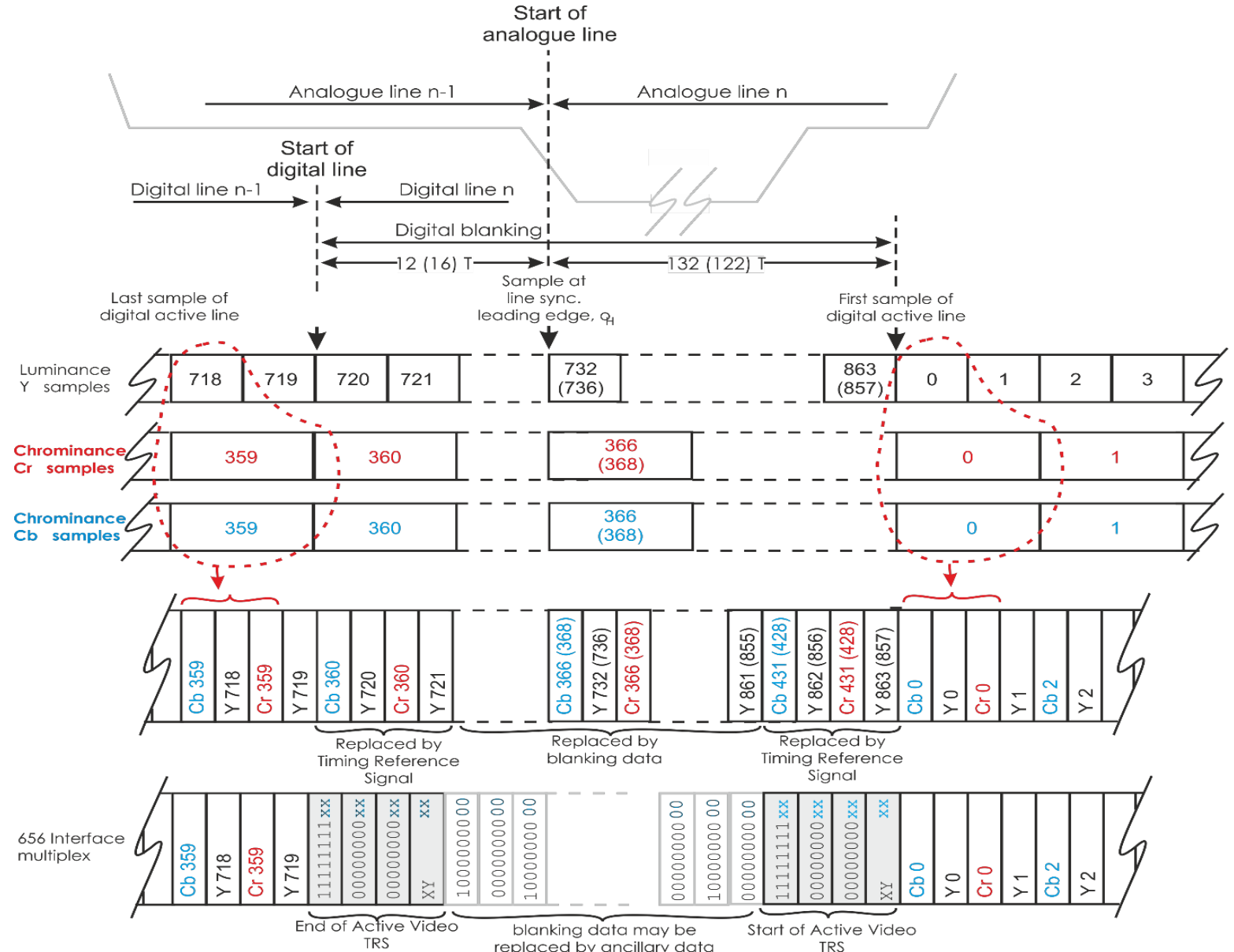
An **SD-SDI (Standard Definition - Serial Digital Interface)** signal, which is part of the **SMPTE 259M** standard used in professional video broadcasting for transmitting uncompressed digital video.

Digital line n represents the entire duration of one horizontal line in the digital domain. This duration corresponds to a fixed number of 10-bit digital words. For 625-line SD systems (like PAL), there are typically **864 10-bit words per digital line**. For 525-line SD systems (like NTSC), there are typically **858 10-bit words per digital line**.

SD-SDI transmits standard definition video (typically 720, encoded digitally). It includes:

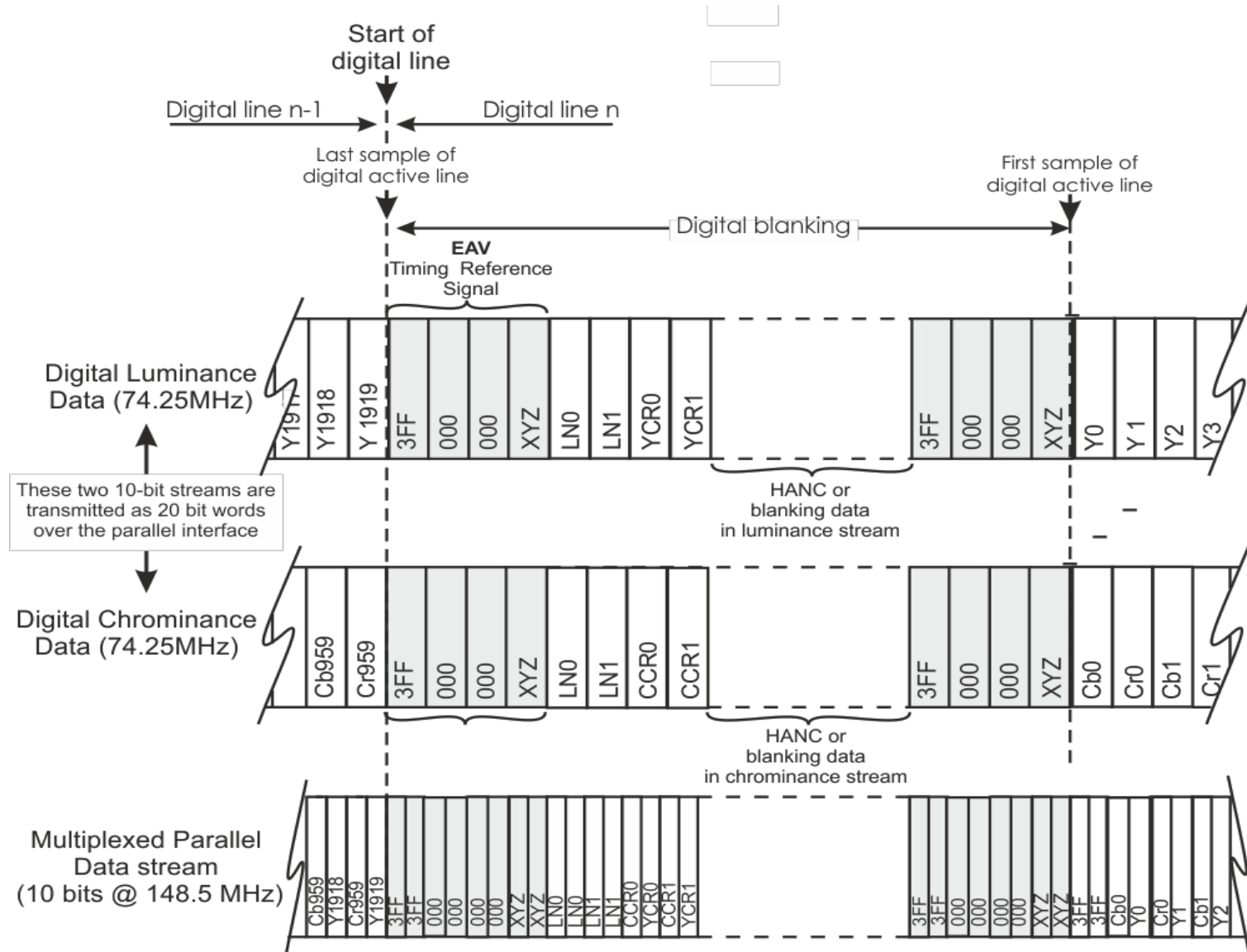
- **Digital video data** (Y, Cb, Cr components)
- **Timing reference signals**
- **Blanking intervals** (digital equivalent of the analog blanking used between lines and frames).

• **Active Video Data:** This section contains the sequence of digital words representing the Y, Cb, and Cr samples for the visible part of the video line. The diagram shows the typical **4:2:2 sampling structure** where for every two Luminance (Y) samples, there is one Cb and one Cr sample.



NOTE: Numbers in parenthesis () are for 525/60 systems
 T = Luminance sample period.
 The two least significant bits of the four word TRS are undefined (shown as xx above), though if generating the signal 11,00,00,00 (binary) are preferred.

- An HD-SDI digital line represents a fixed number of 10-bit words transmitted serially over time.
- LN0, LN1:** Line Number data. **These HANC packets carry the digital line number** (e.g., 1 to 1125 in a 1080/59.94i system). This is crucial for the receiver to know which line of the frame it's processing.
- YCR0, YCR1:** Luminance check words. These HANC packets contain error **detection/correction codes** calculated from the Luma pixel data of the line.
- Multiplexed Cb/Cr stream** running at the **same word rate as the Luma stream**.
- Multiplexed Parallel Data Stream (10 bits @ 148.5 MHz):** This is the stage where the independent Luma stream (with its TRS/HANC) and the multiplexed Chrominance stream (with its TRS/HANC) are combined into a single data path.



- **Active Lines** counts for different standards: 1080 (for 1080 HD), 720 (for 720p HD), 576 (for PAL SD), and 480 (for NTSC SD).
- **Total Lines** counts for the corresponding standards (1125 for 1080 HD, 750 for 720p HD, 625 for PAL SD, 525 for NTSC SD).
- E.g., Out of 1125 total lines per frame, only 1080 carry active picture. The remaining $1125 - 1080 = 45$ lines (spread across the top and bottom of the frame) are part of the **Vertical Blanking Interval (VBI)**.

Vertical Ancillary Data (VANC) - Timecode, Subtitles etc.



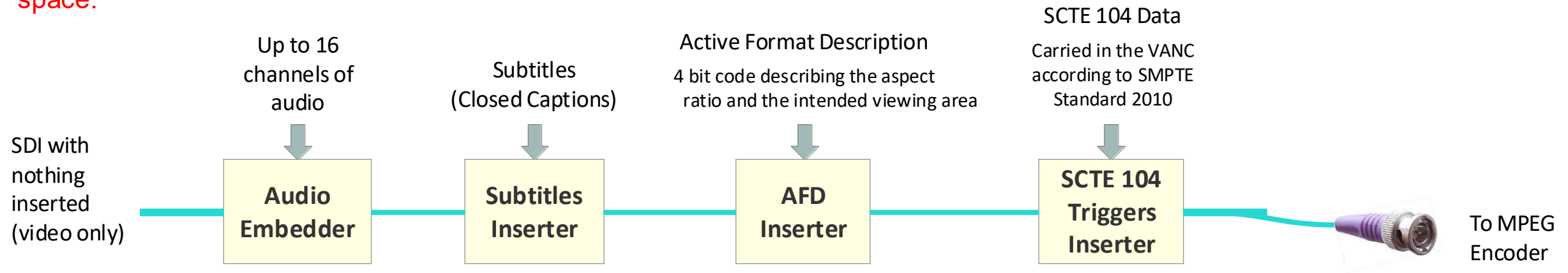
Horizontal Ancillary Data (HANC) - Embedded Audio Data

Active Lines
1080 / 720 / 576 / 480

Total Lines
1125 / 750 / 625 / 525

VANC and HANC regions are the "extra space" where essential non-picture data like embedded audio, timecode, and metadata are embedded.

Each block represents a function or device that formats and inserts a specific type of data into the available HANC or VANC space.



- **Ancillary data** for SDI is defined in **SMPTE ST 291M**.

- Standard governs the structure, format, and identification of ancillary data packets within an SDI stream. It defines how different types of data are packaged with headers, data words, and checksums so that receiving equipment can correctly identify, extract, and interpret the different types of embedded information.

- Each data type has a code which allows equipment to know what kind of data is contained within a particular ancillary data packet - SMPTE maintains a register of the codes.

- <https://smpte-ra.org/smpete-ancillary-data-smpete-st-291>

- **HANC – Horizontal Ancillary Data** – in the time gaps between the lines.

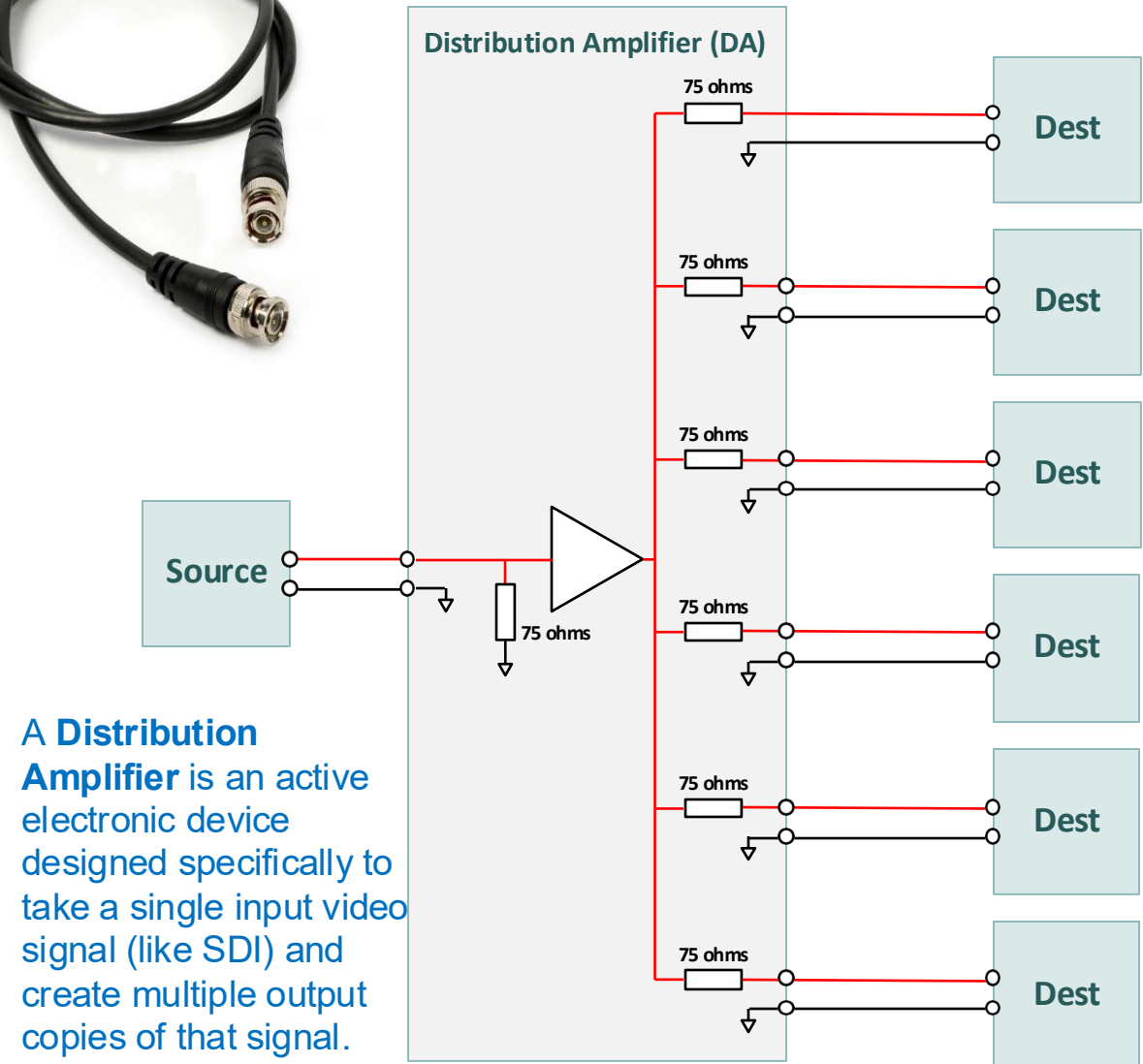
- **Embedded Audio** - takes external digital audio signals and formats them into ancillary data packets according to the standard for embedded audio in SDI (specifically SMPTE ST 299-1 for embedded audio).

- Sometimes called **VBI** data - **V**ertical **B**lanking **I**nterval.
- **EDH** – Error Detection and Handling (for SD SDI only. CRC check is built into HD-SDI).
- **VPID** – Video Payload Identifier
 - Increasingly being used to describe the video format.
 - In early SDI it was possible to determine the video format by counting lines and samples
 - With the introduction of dual link interfaces, this is no longer possible
 - VPID is defined in SMPTE 352M
 - A list of the codes is also contained in the SMPTE Register:
 - <https://smpte-ra.org/video-payload-id-codes-serial-digital-interfaces>
 - VPID is crucial for automated systems and interoperability. Equipment can read the VPID packet in the VANC to instantly know the format of the incoming video stream without needing to rely on timing analysis or manual configuration.
- **Other SMPTE 291M Ancillary Data**
 - Timecode, closed captions, triggers (See SCTE 104), Lip sync data, control data, camera metadata etc...

Standard	Name	Introduced	Bitrates	Example Video Formats
SMPTE 259M	SD-SDI	1989	270 Mbit/s	480/60i, 576/50i
SMPTE 292M	HD-SDI	1998	1.485 Gbit/s	1080/60i, 1080/59.94i, 1080/50i, 1080/30p, 1080/29.97p, 1080/25p, 1080/24p, 1080/23.98p, 720/60p, 720/59.94p, and 720/50p
SMPTE 372M	Dual Link HD-SDI	2002	2.970 Gbit/s	1080/60p, 1080/59.94p, and 1080/50p, as well as 4:4:4 encoding, greater colour depth, RGB encoding, alpha channels, and nonstandard resolutions
SMPTE 424M	3G-SDI	2006	2.970 Gbit/s	
	Quad link HD-SDI		5.940 Gbit/s	2160/25p, 2160/30p (4:2:2)
SMPTE 2081-10	6G-SDI	2015	5.940 Gbit/s	
	Quad Link 3G SDI		11.88 Gbit/s	2160/50p, 2160/60p (4:2:2)
SMPTE 2081-11	Dual link 6G-SDI	2016	11.88 Gbit/s	
SMPTE 2082	12G-SDI	2015	11.88 Gbit/s	
SMPTE 2081-12	Quad Link 6G-SDI	2016	23.76 Gbits/s	2160/50p (4:4:4), 2160/60p (4:4:4), 2160/100p (4:2:2) 2160/120p (4:2:2), 4320/25p (4:2:2), 4320/30p (4:2:2).
SMPTE ST-2083	24G-SDI	-	23.76 Gbits/s	

- For **quad SDI** there are **2 ways** of splitting the data.
 1. **SQD (Square Division)**
 - Square Division picture divided into 4 quadrants. each quadrant sent down a link.
 2. **2IS (2 Sample Interleave)**
 - Odd lines: First 2 samples of the line on link A, next 2 samples link B etc.
 - Even lines : First 2 samples of the line on link C, next 2 samples link D etc.

- **SDI** (Serial Digital Interface) and **ASI** (Asynchronous Serial Interface) use **coax cable** with **BNC connectors**.
- The cable must be terminated in **75 ohms**: **Every cable designed for high-frequency signal transmission has a specific characteristic impedance**. For standard video cable, this impedance is 75 ohms. Proper **termination impedance** at the ends of the cable matches the cable's characteristic impedance (75 ohms).
- A passive split disrupts the 75-ohm impedance matching. When you split a cable this way, the signal sees a different impedance than 75 ohms at the split point and at the inputs of the receiving devices. This causes reflections and severe signal degradation to all connected devices.
- If you need **multiple copies** of a signal and there is only one output available, you need to use a **Distribution Amplifier (DA)**.



DECS

DV CTP
108.A.

AXON DA
108.CH1.

Please ignore the strikethrough labels, they are not correct yet.

OP A1 / OP A2 – JF 108.G/I.1 – N/C / N/C (for monitoring only)

OP A3 > JF 108.E.1

OP B1 > N/C, OP B2 > N/C (temp cables to DCS21 114)

OP B3 > Evertz RTR IP 61 [078 FNSA]

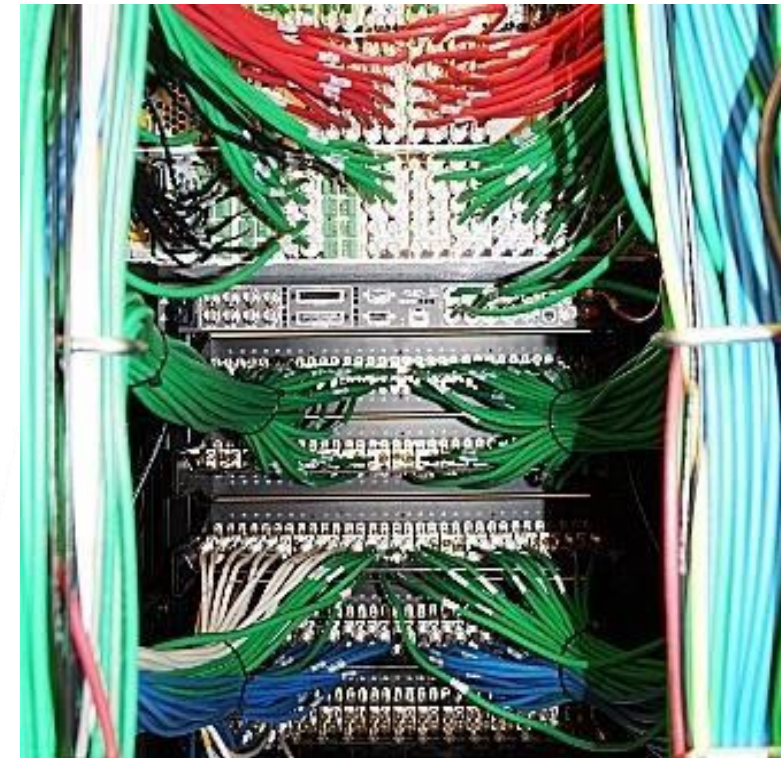
CTP = Central Termination Panel: passive panels used for organizing and managing large numbers of coaxial cables.

CTPs provide flexibility for patching and routing signals, while the DAs provide the necessary signal splitting and isolation for reliable distribution.



1U 1x12 BNC : 1 rack unit high panel with one row of 12 BNC connectors.

An DA made by Axon



- BNC (75Ω)

- Bayonet Neill–Concelman
- **Locking.** Tool available for dense BNCs (e.g. on the rear of routers)

- MUSA

- **Multiple Unit Steerable Array**
- Developed for RF aerial connections – **Therefore 50Ω**
- **Not locking** - can be pulled out accidentally (like HDMI plugs).
- Adopted for use in **video jackfields** (easy to slide in and out).
 - Impedance mismatch not critical for analogue video (compared to the requirements of digital signals.).
- **Original MUSA not suitable** for digital video (due to impedance mismatch) – especially **>3G SDI**.
- **New**, reengineered versions are available
 - Which are **75Ω** and therefore **OK** for 3G+ SDI (MUdigSA)



- As digital signals travel down a cable, they lose strength and their timing can become distorted.
- This degradation increases with distance and with the frequency of the signal.
- These cables vary in characteristics, primarily in their physical thickness (gauge) and construction, which affects their signal carrying capabilities.
- As the SDI bitrate increases, the maximum transmission distance decreases significantly for the same cable type.
- This dramatic reduction in distance is because **higher frequency components** are required to transmit data at higher speeds. Cables cause greater signal loss (attenuation) at higher frequencies.

	Standard	259	292M	292M
	Name	SDI	HDSDI	3GSDI
	Bit rate Mbit/s	270	1485	2900
	Belden Cable			
Yellow series	179DT	116m	34m	24m
	1855A	241m	79m	46m
Blue series	1505F	261m	69m	
	1505A	338m	94m	67m
Green series	1694A	436m	122m	82m
	7731A	622m	168m	110m

- **SMPTE 297-2006** defines an optical fibre system for transmitting **SDI** signals or **ASI**.
- Fibre connections are needed for HD-SDI connections of more than about 200m (coaxial cable has significant signal loss, especially at the high frequencies and higher bitrates).
- Optical signals experience significantly less attenuation (signal loss) over long distances compared to electrical signals on copper cables, especially at high frequencies/bitrates. This allows SDI signals to be transmitted reliably over much greater distances – typically several kilometres, and sometimes tens of kilometres.
- This **not** IP / Ethernet. Pure SDI over fibre. The data is transmitted as light pulses.
- In optical form SDI signals can be split using a passive optical splitter instead of a distribution amplifier (DA).

SFP (Small Form-factor Pluggable)

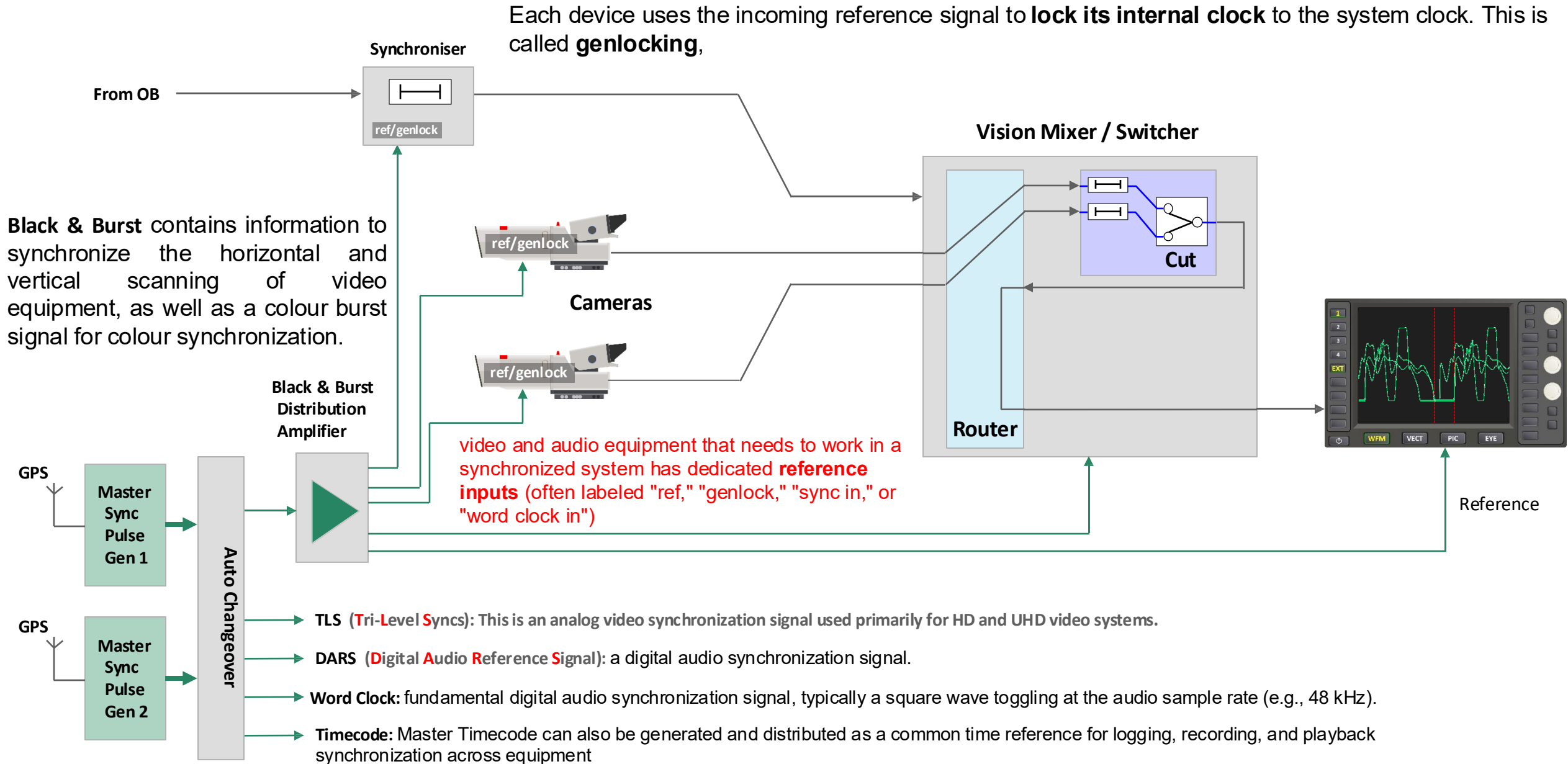


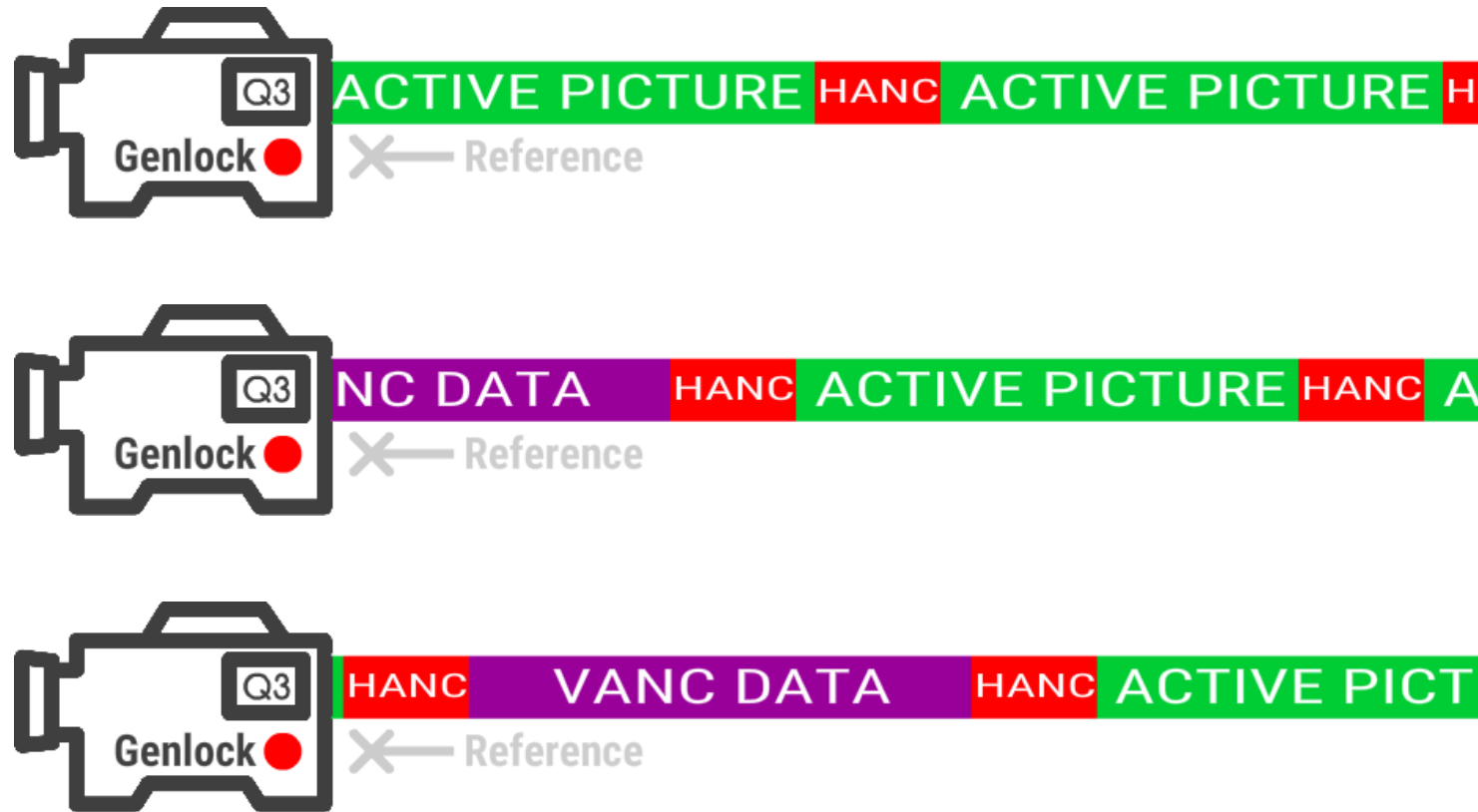
Introduction to Video Timing

- Synchronization ensures that all elements of a video program (pictures, sounds, graphics, etc.) are aligned with each other.
- Audio to video sync (**lip sync**).
 - More than 1 frame of error starts to be noticeable.
 - Worse if audio is ahead of the video.
- Audio sample sync – to ensure **clean switching** and mixing of **audio** signals.
 - Must be phase and frequency locked.
- Video sync – to ensure **clean switching** between **video** sources.
 - Must be phase and frequency locked.
 - Switching must occur during field or frame blanking in the video.
- Timecode – to **mark and identify** video frames when editing.
 - Timecode assigns a unique digital address (in the format HH:MM:SS:FF, e.g., 01:23:45:10 for Hour:Minute:Second:Frame) to every single frame of video. This is invaluable in post-production for navigating through footage, precisely identifying specific moments, setting edit points, and managing large amounts of media.

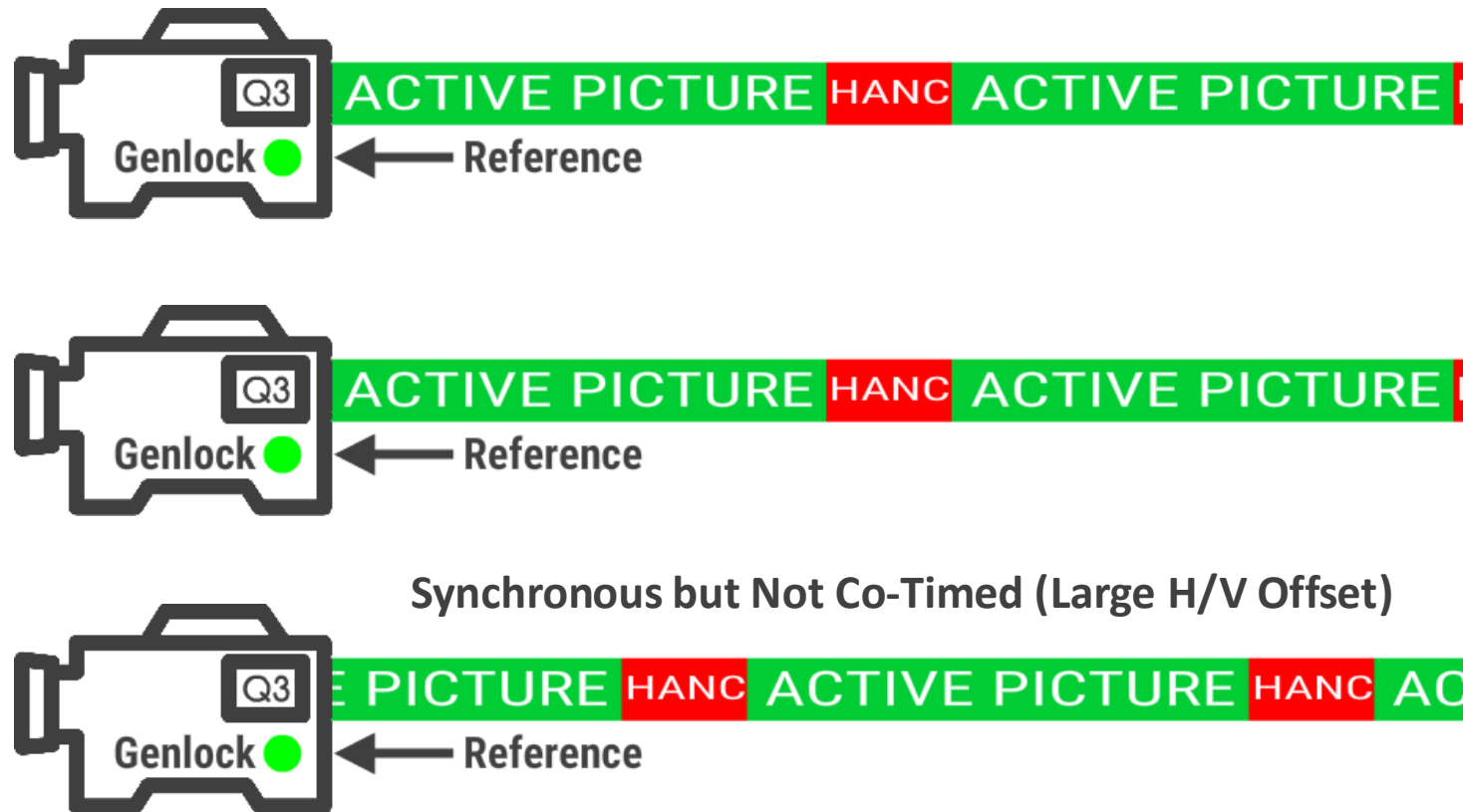


A timecode reader





- Every piece of digital video hardware has its own internal clock, these internal clocks aren't perfectly stable and need to be locked to a common external reference to ensure all equipment operates in precise synchronization.
- Genlock input shown on the devices in the diagram is specifically designed to receive an **external timing reference signal**.
- When a device receives this external reference signal, its internal timing circuitry uses the reference to **control and discipline its internal oscillator**. It adjusts its own clock frequency and phase to lock onto the incoming reference signal.



H/V Offset

Adjust output timing relative to reference:

- **V (Vertical)** Coarse adjust
- **H (Horizontal)** Fine-tune

- Video signal is running at the **correct frequency and frame rate**, locked to the master timing reference signal.
- Even though signals might be synchronous (running at the same speed), they might **not be co-timed**.
- They are running in parallel at the same speed, but one is slightly ahead or behind the others.

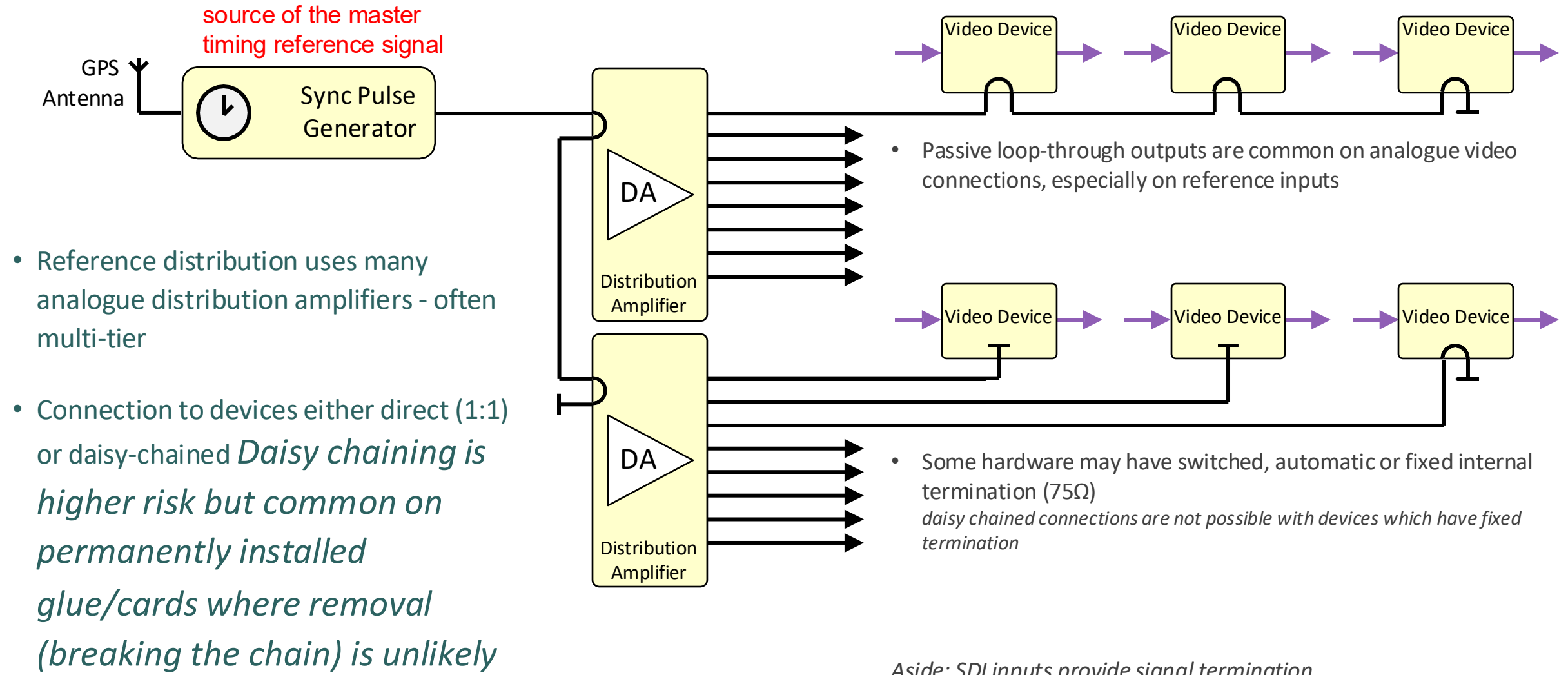


H/V Offset

Adjust output timing relative to reference:

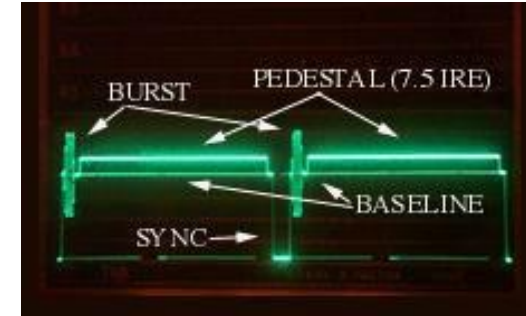
- **V (Vertical)** Coarse adjust
- **H (Horizontal)** Fine-tune

Genlock reference signal needs to be sent to potentially many different pieces of video equipment

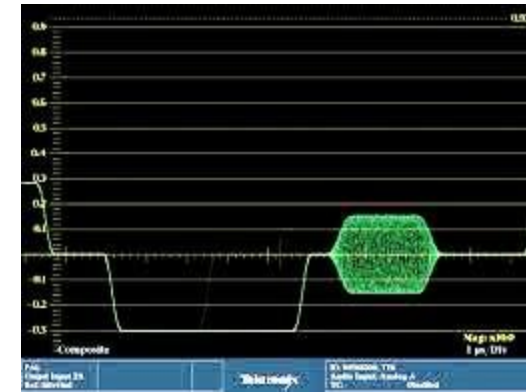


Aside: SDI inputs provide signal termination
'SDI Loop Out' is active (buffered/re-clocked)

- Black & Burst (USA: Blackburst)
 - A standard definition, analogue composite (PAL or NTSC) signal with a black picture.
 - Consists of line and field syncs and the colour burst.
 - Used as a reference for all types of system (even HD digital ones!)
 - All required timings can be derived from it.
- Tri-Level Syncs
 - A high definition, monochrome, analogue signal with a black picture.
 - Sometimes used as the reference in an HD digital system.
- Word Clock
 - 3V to 5V p-p square wave at the audio sampling frequency, carried over 75 ohm coax.
 - Used to lock audio equipment.
- Timecode
 - A digital signal carrying the time as **hours:minutes:seconds:frames**
 - May be carried in different ways - see next slides.



NTSC black & burst



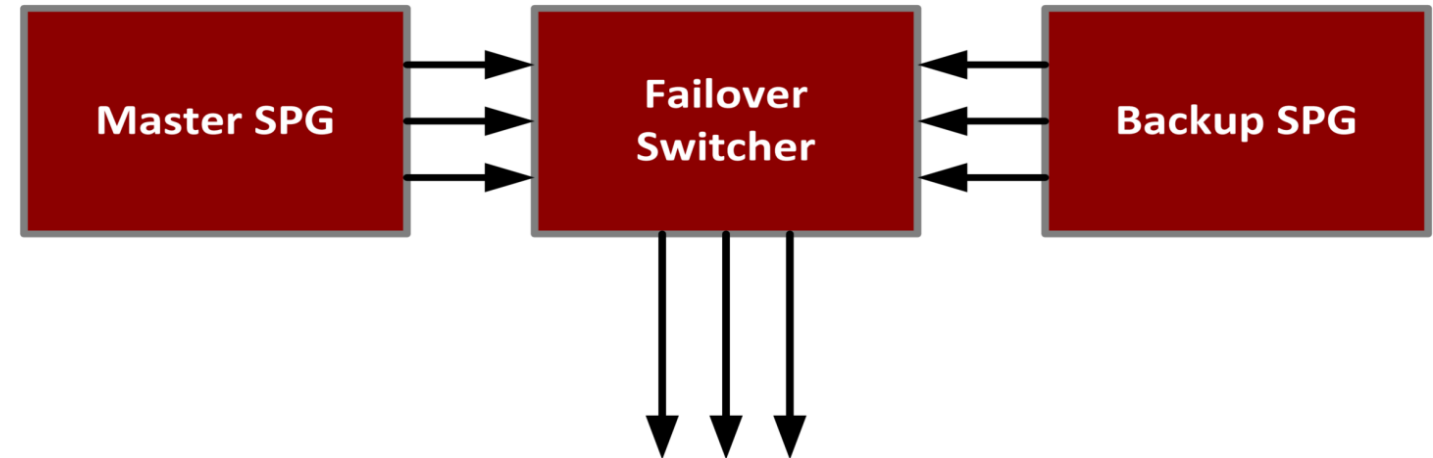
Expanded view of the sync and burst from PAL black & burst



Expanded view of a tri-level sync pulse

- Sync Pulse Generator (SPG) Outputs.

- Black and Burst (B&B).
- SDI / HD-SDI.
- Tri-Level Syncs (TLS) for HD.
- Word Clock.
- Test Signals.



- In critical applications like broadcast, SPGs are often deployed in redundant Master/Backup configurations with an automatic failover switcher to ensure uninterrupted timing reference distribution.



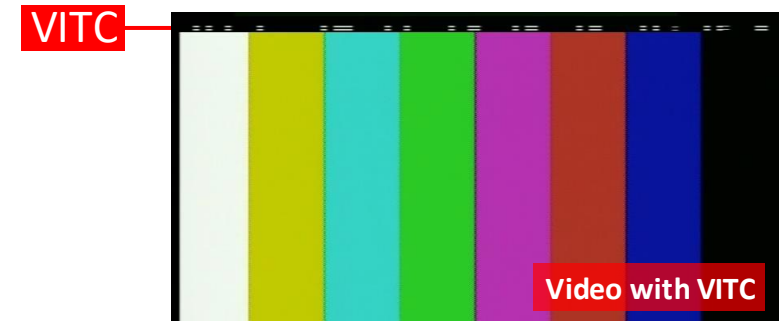
- Timecode is an important element in video production and post-production for identifying and synchronizing individual video frames.
- a system of assigning a unique address to every single frame of video
 - The standard timecode format is **HH:MM:SS:FF**
 - **Hours:Minutes:Seconds:Frames**
- Timecode counts frames sequentially. The range of the "Frames" field depends on the video's frame rate.
 - **00:00:00:00** to **23:59:59:24** in **25 FPS** systems.
 - **00:00:00:00** to **23:59:59:29** in **30 FPS** systems.
- Timecode can be **carried**:
 - As a signal in the **vertical interval of analogue video (between frames)**.
 - Vertical Interval Timecode (**VITC**).
 - As a **digital ancillary data** in an **SDI signal**.
 - Digital Vertical Interval Timecode (**D-VITC**).



A timecode reader



An Old Leitch Studio Clock



- TOD – Time of Day
- TOD timecode is when the timecode generator is set to output timecode that matches the actual clock time of day.
 - Used for:
 - Studio / broadcast centre distribution
 - Camera recordings
- Master tape / final edit
- A standardized timecode structure is used on master tapes or final edited files to provide consistent cueing and identification for broadcast and distribution.
 - Usually has a standard timecode, for example:
 - The programme starts at 10:00:00:00. (to provide a common reference point.)

- **Drop Frame Timecode** is a method used to make the timecode clock more accurately match real clock time in video systems running at 29.97 frames per second.
- **"30"FPS** video is normally **29.97FPS**
 - Really **$30 \times 1000 / 1001$** or **29.97002997002997002997002997002997.... FPS**
 - **0.1%** discrepancy between **time code time** and **'real time'**
 - Creates **3.6 second per hour** error.

Over longer programs or recordings, this cumulative error becomes significant, meaning the timecode value displayed does not accurately reflect the actual duration of the program or recording based on a real clock.

- **Drop frame time code** corrects discrepancy and make the timecode clock run at the same rate as real clock time.
 - Skips first two frames of the first second of every minute.
 - Except every tenth minute.

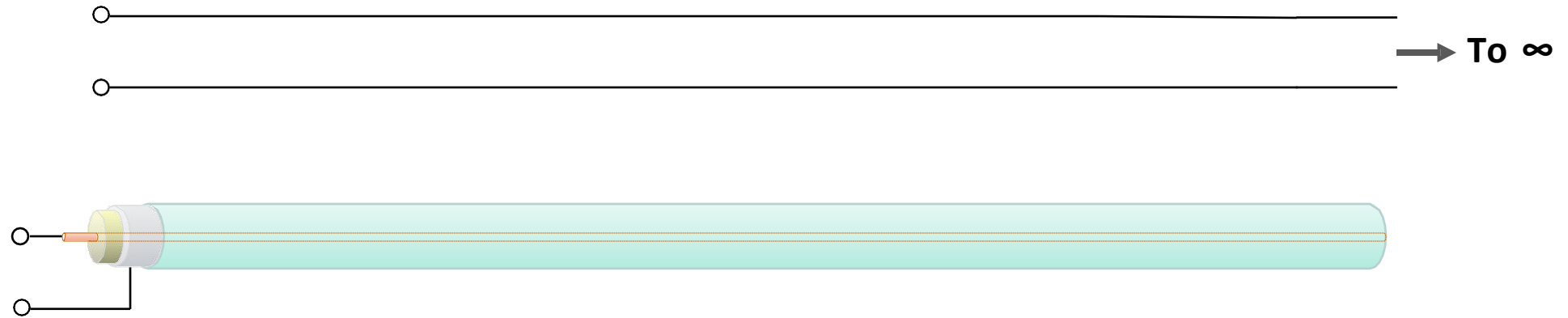
- **Burnt-in Timecode** is a visual display of the timecode value that is permanently superimposed onto the actual video image itself. It's not carried as separate data or metadata; it's part of the visible picture information.
- Burnt-in Timecode is a textual overlay of the timecode value (HH:MM:SS:FF) placed directly onto the video frame.
- The timecode value changes frame by frame to show the timecode address of the current frame.

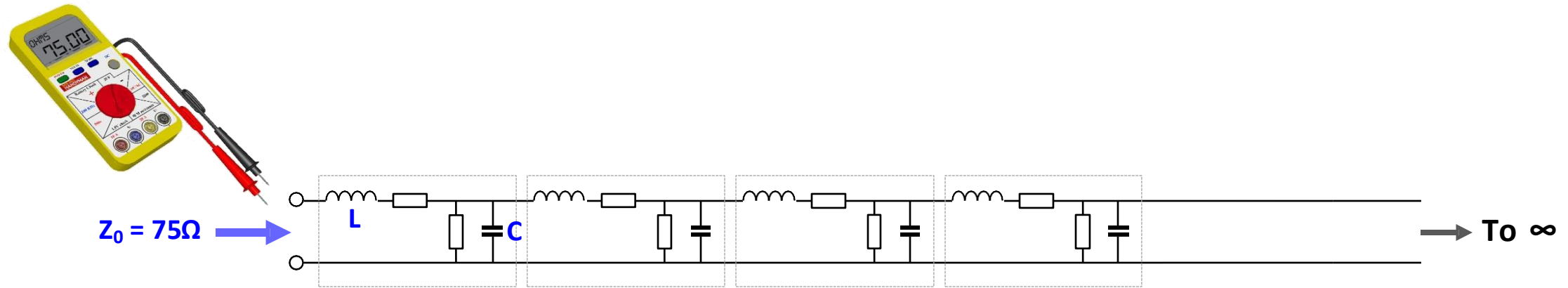


The End

Questions?

Cables and Terminations

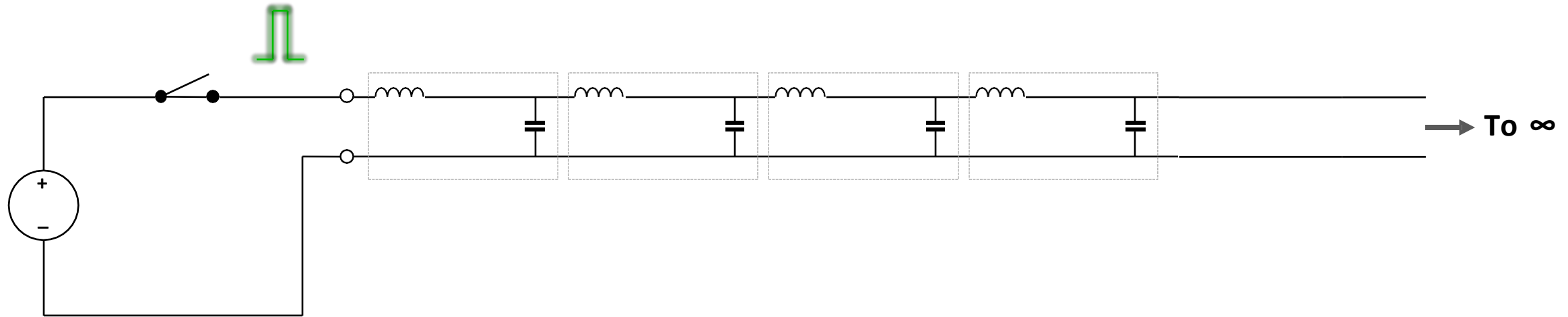




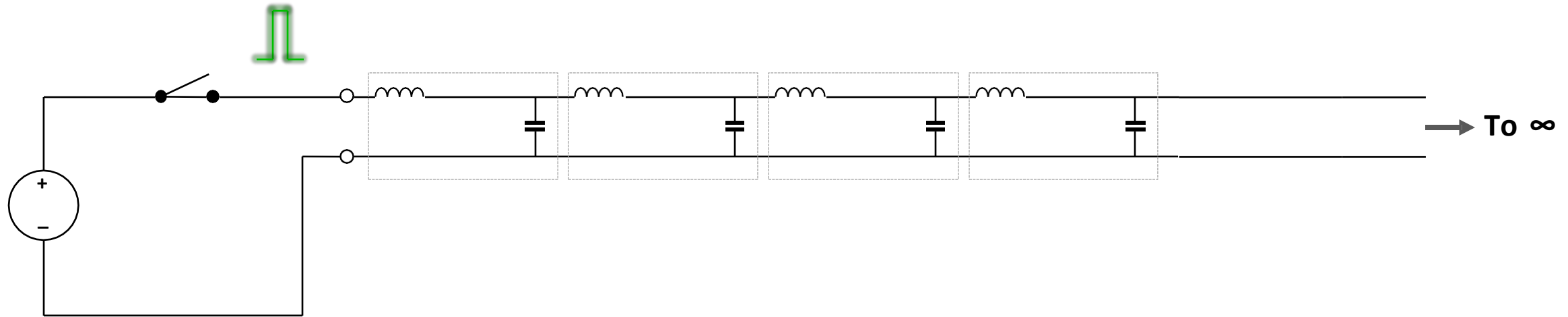
Characteristic Impedance:

$$Z_0 = \sqrt{\frac{L}{C}}$$

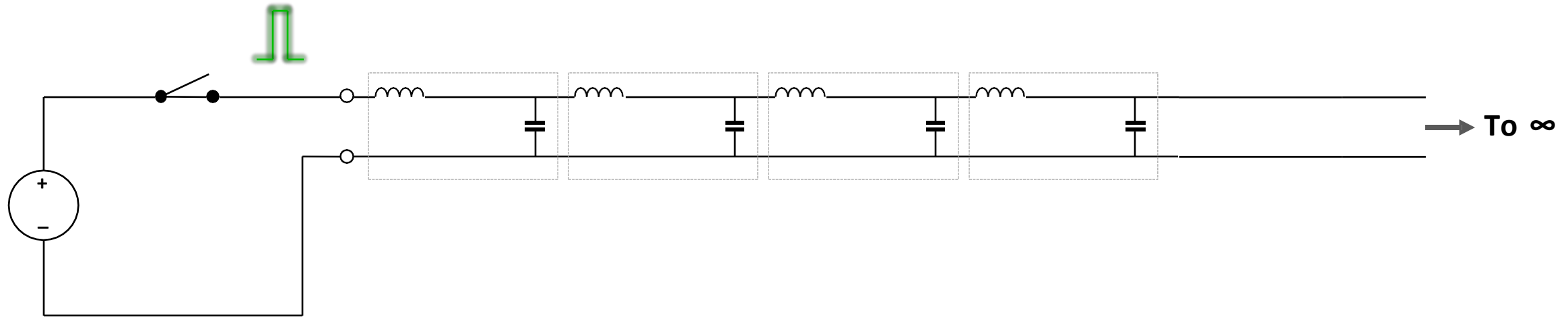
1

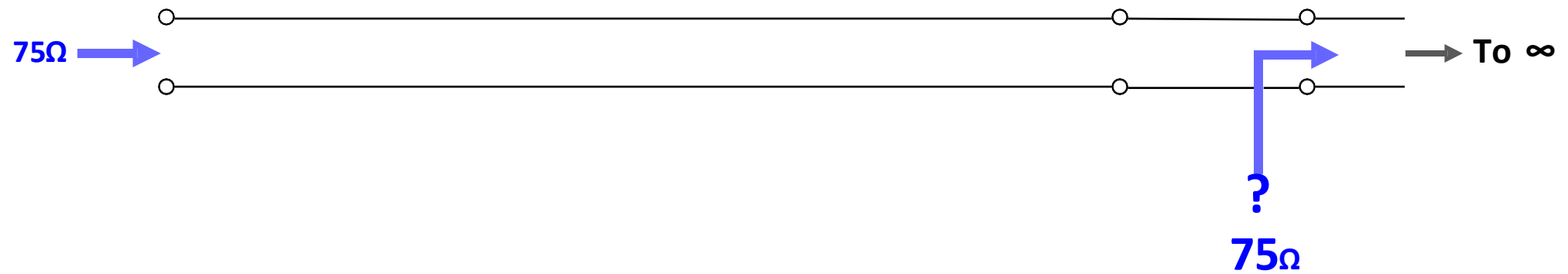


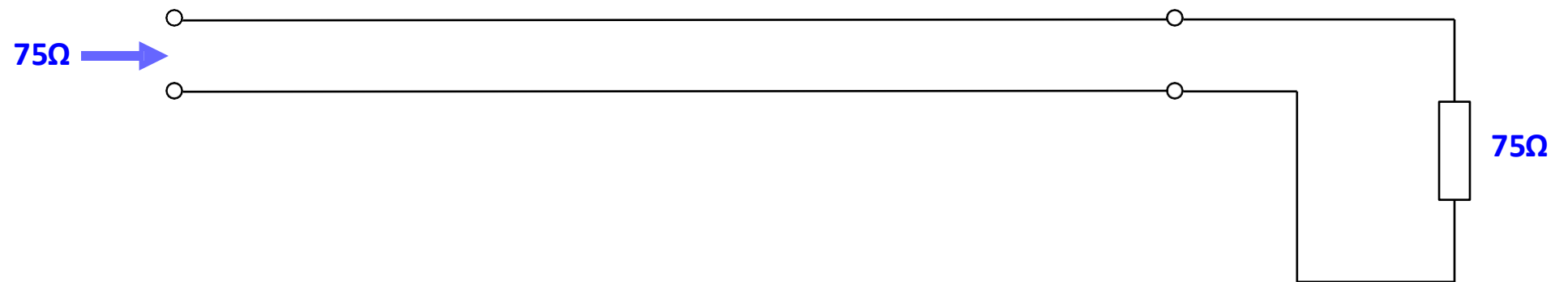
2

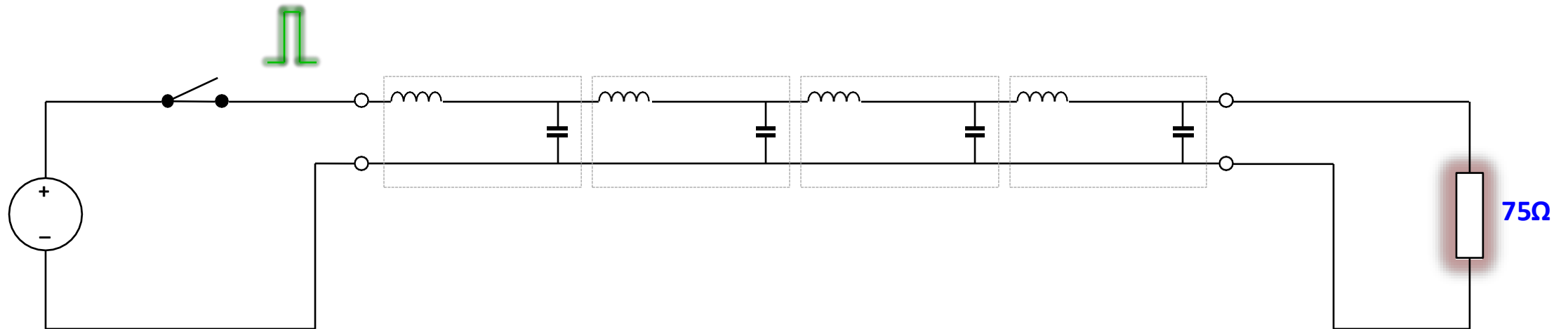


3

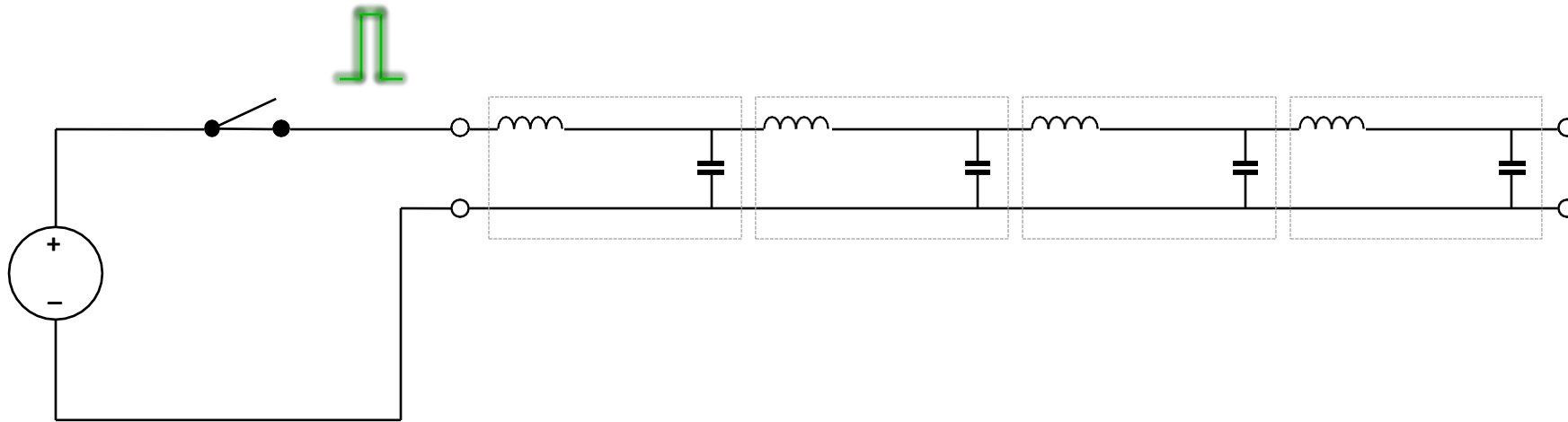




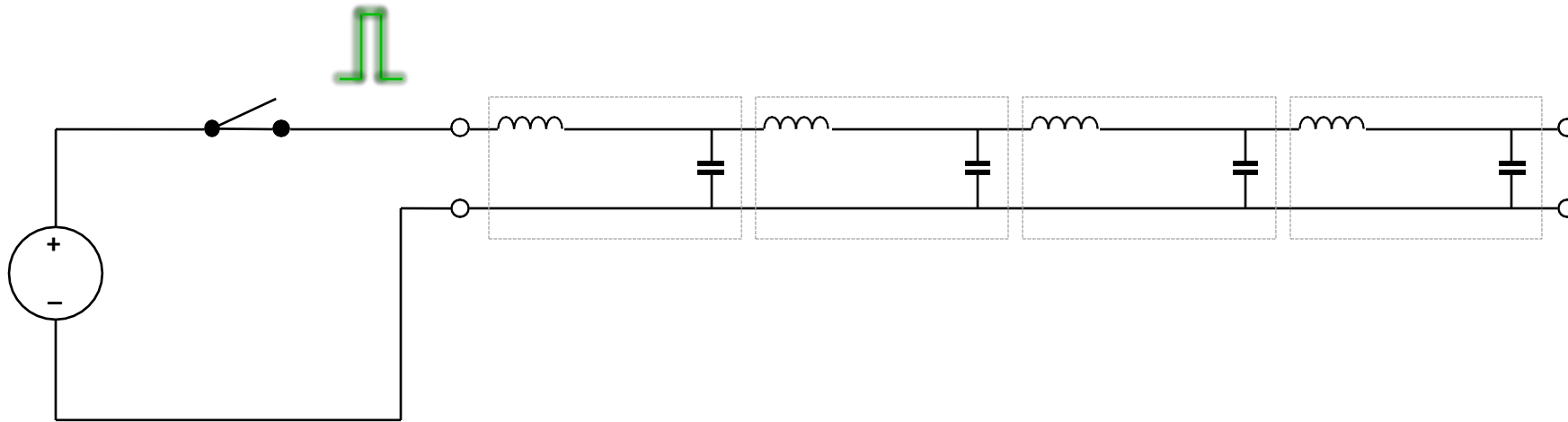




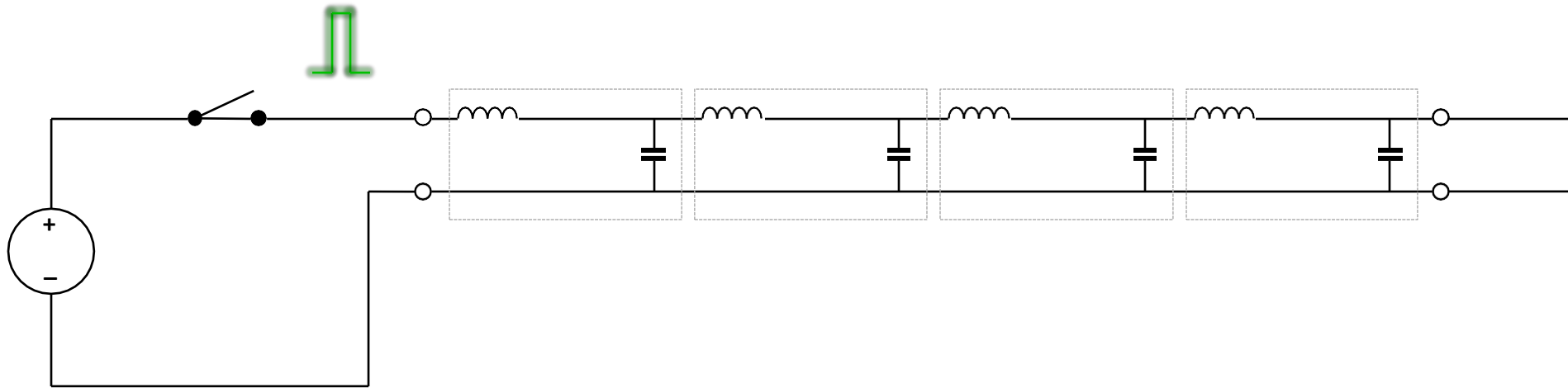
Correct Termination



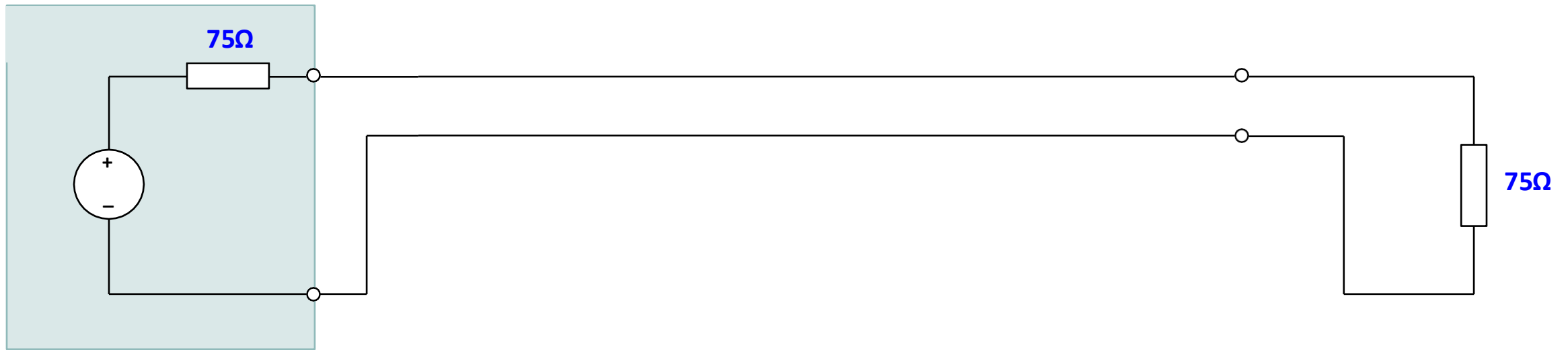
Open Circuit Termination

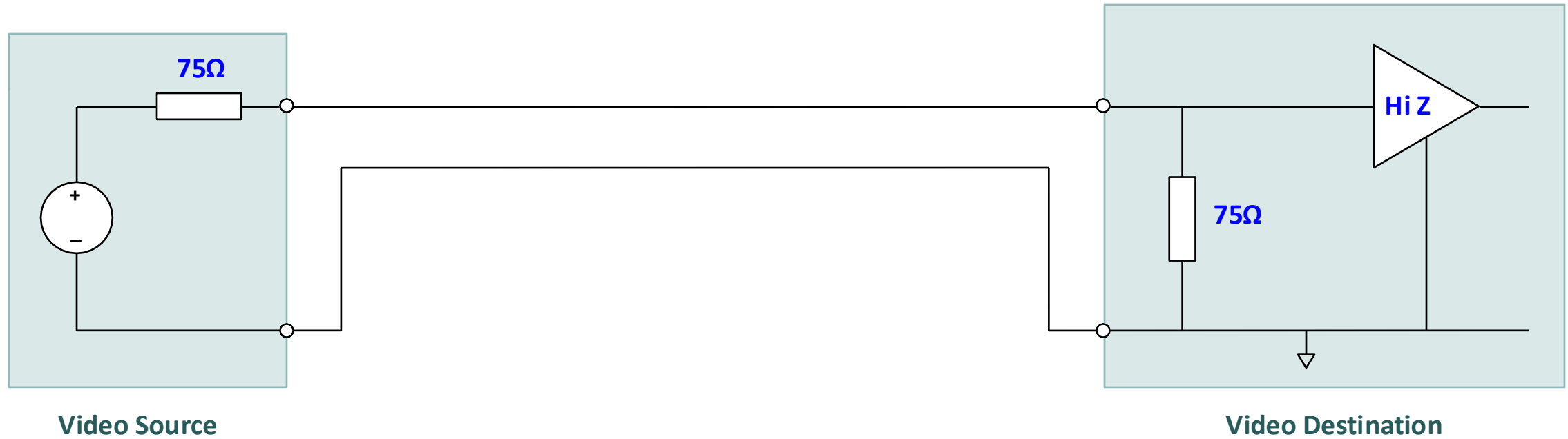


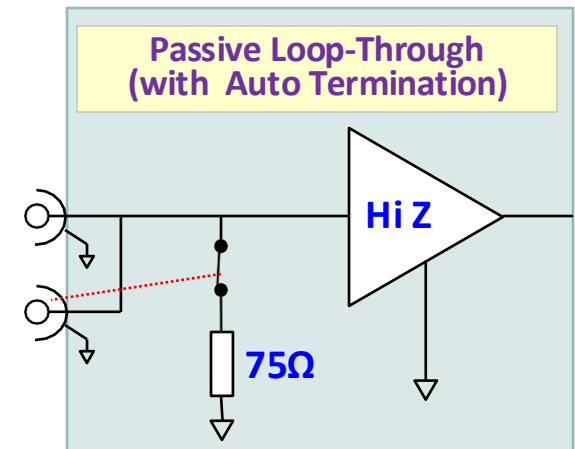
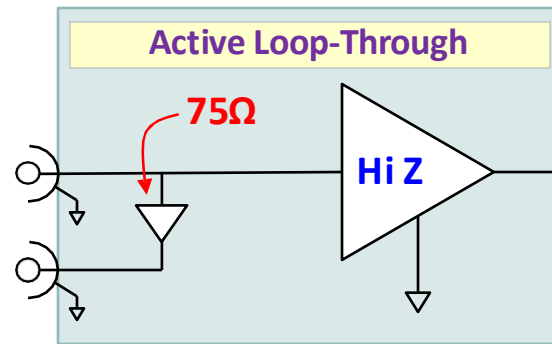
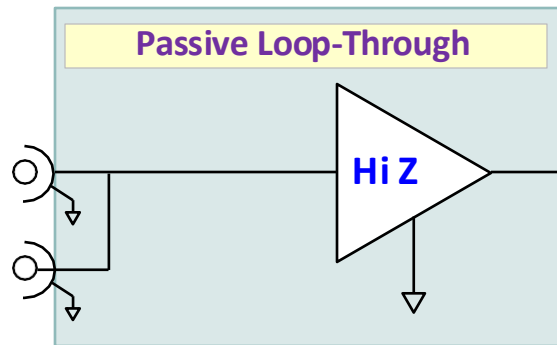
Open Circuit Termination

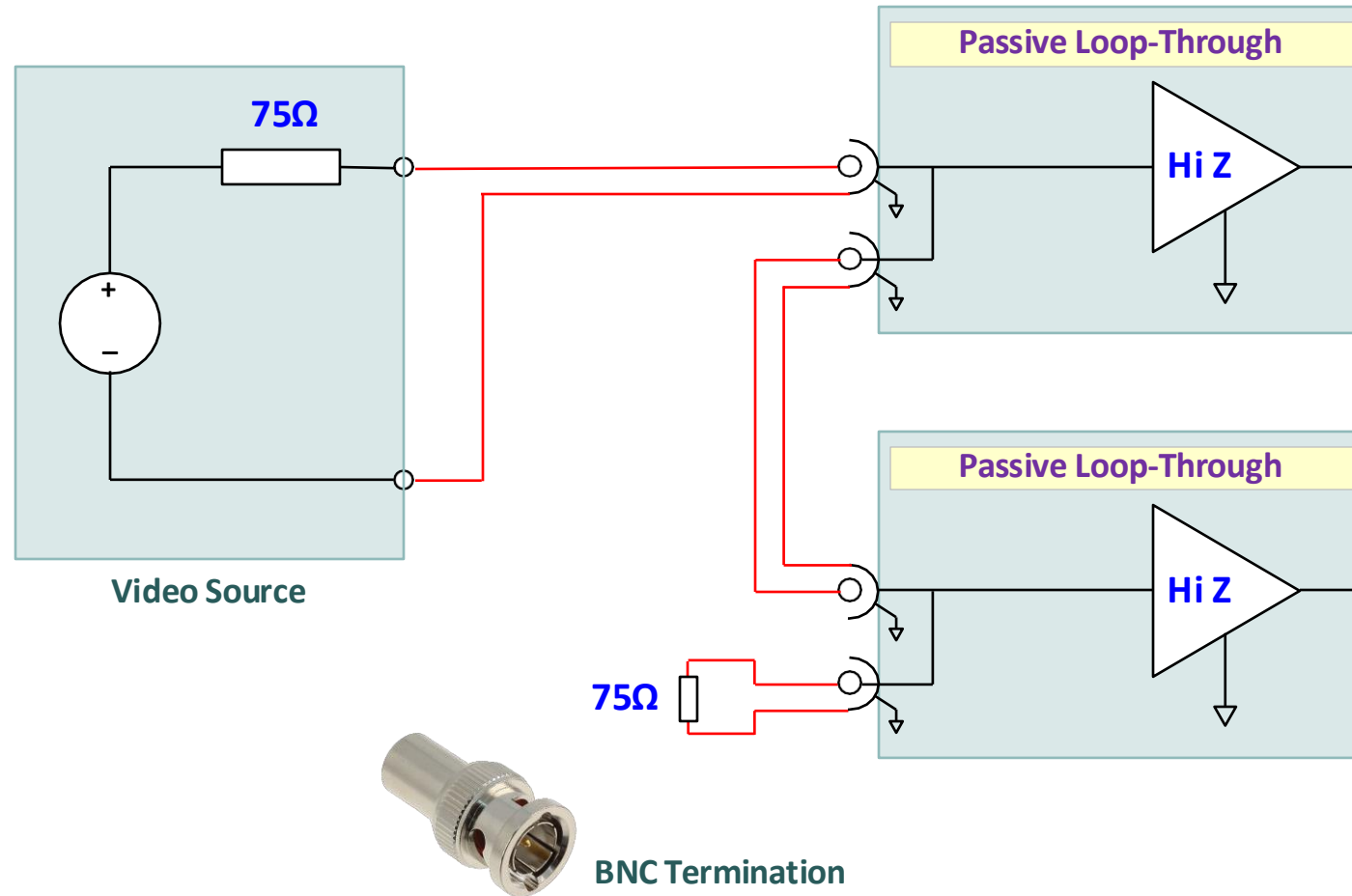


Short Circuit Termination









$$\frac{720}{2} = 360 \text{ ns}$$

$$S = \frac{D}{T}$$

$$D = S \times T$$

$$360 \times 10^{-9} \times 3 \times 10^8$$

$$= 108 \text{ m}$$